

CONDUCTOR REDUCTION AND PRIMARY STRUCTURES OF MULTIPLICATION FAILURE SCHEMES

QIAN QI

ABSTRACT. For binary subseries containing the sections vanishing on a finite divisor, we identify the higher-multiplication cokernel with a finite-algebra cokernel, in families and after arbitrary base change. For generating contact pencils this gives the entire failure scheme in every degree. Above the contact-length threshold the ideal stabilizes and has an explicit primary decomposition for every multiplicity partition of the divisor. We determine all coranks, exact nilpotency indices, successive nilradical quotients, and local weighted-arrangement equations. Although fixed-contact fibres can be highly nonreduced, the moving-contact failure divisor is integral. Its normalization is the smooth finite incidence marking a length-two subdivisor not separated by the pencil; an explicit differential criterion determines its singular support, including collisions. These results apply to subseries of arbitrarily large codimension. We also retain the quadratic binary component classification, polar residual equations and wall singularities, and the all-degree hyperplane theorem with its embedded evaluation curve. Complete information-recovery and statistical applications are reproduced in separate appendices.

1. INTRODUCTION: MULTIPLICATION THROUGH A FINITE CONTACT ALGEBRA

Let $V_n = H^0(\mathbb{P}^1, \mathcal{O}(n))$. A proper subseries $U \subset V_n$ may fail to generate the complete series in higher symmetric degree even when it has no base points. We study the *scheme* of this failure: it is defined by the zeroth Fitting ideal of the cokernel of $\mathrm{Sym}^m U \rightarrow V_{mn}$. The scheme remembers multiplicities which neither its support nor a tangent calculation determines.

Our first object is a subseries containing all sections vanishing on a prescribed finite divisor. Let D have length d , put $W_D = H^0(\mathcal{O}(n)(-D))$, and let

$$A \subset E_{n,D} := H^0(\mathcal{O}_D(n)), \quad U_A = \mathrm{res}_D^{-1}(A).$$

We require A to generate $\mathcal{O}_D(n)$; repeated points of D are allowed. These are entire open relative Grassmannians of subseries containing W_D , not selections of failure points. For $n \geq 2d - 1$ we prove an isomorphism of coherent cokernel sheaves

$$\mathrm{coker}(\mathrm{Sym}^m \mathcal{U}_A \rightarrow V_{mn}) \simeq \mathrm{coker}(\mathrm{Sym}^m \mathcal{A} \rightarrow E_{mn,D}) \quad (m \geq 2). \quad (1.1)$$

It holds over the moving-divisor parameter space and after arbitrary base change. Thus all Fitting schemes, including nonreduced fibres, are transported to a multiplication problem on a finite algebra. The point is a global surjectivity statement for products containing W_D , not an elimination of an invertible block at a single point.

For pencils, the finite problem has a complete solution. Write $G_D^\circ \subset \mathrm{Gr}(2, E_{n,D})$ for the generating locus. On a smooth frame chart choose a basis a, b with a a unit on D , and put $v = b/a$. The cokernel is presented by $[1, v, \dots, v^m]$ in the length- d algebra $H^0(\mathcal{O}_D)$. For $m \geq d - 1$ the failure ideal is independent of m .

Date: September 22, 2026.

2020 Mathematics Subject Classification. 14M12, 14H50, 13C40, 14B05, 14N15.

Key words and phrases. Multiplication of sections, finite contact algebras, primary decomposition, normalization, nilpotent structures, polar residual spaces.

Theorem 1.1 (Contact-pencil failure schemes). *Let $d \geq 4$, $n \geq 2d - 1$, and $m \geq d - 1$. The maximal-minor failure scheme on G_D° is a Cartier divisor. If $D = \sum_i d_i p_i$, its complete primary decomposition is*

$$\mathcal{J}_m = \bigcap_{i: d_i \geq 2} \mathcal{I}_{T_i}^{\binom{d_i}{2}} \cap \bigcap_{i < j} \mathcal{I}_{E_{ij}}^{d_i d_j}. \quad (1.2)$$

Here T_i is the divisor on which A does not separate $2p_i$, and E_{ij} is the divisor on which it does not separate $p_i + p_j$. These are distinct smooth irreducible divisors. There are no embedded associated points. On smooth frame charts the exact local equation is a weighted product of linear forms; its weights, nilpotency indices, successive nilradical quotients, tangent cones, and every multiplication corank are given in Theorems 3.2 and 3.3. Corollary 4.1 determines all components and their primary multiplicities over every moving multiplicity stratum.

Allowing D to move, the failure divisor on the relative generating Grassmannian is integral. Its normalization is smooth and is the finite incidence which marks a length-two subdivisor $Z \leq D$ not separated by A . Theorem 4.2 gives this normalization, and Proposition 4.3 gives an exact test for its singular support.

These subseries have codimension $d - 2$ in V_n , which is unbounded as d varies. In contrast to the hyperplane family, their scheme structure stabilizes once $m \geq d - 1$. It nevertheless varies sharply with the multiplicities of D : for $D = dp$ the fixed-divisor failure scheme is the $\binom{d}{2}$ -fold ramification divisor, whereas for a reduced divisor it is a reduced arrangement of pair-identification divisors. Both are fibres of the same integral moving-divisor scheme. The normalization explains why nonreduced fixed-contact fibres must not be identified with embedded components of the total family.

The proof has three steps. A base-point-free pencil argument fills all products vanishing twice on D ; the first normal quotient then fills all products vanishing once. This proves (1.1). A power-basis determinant in the finite algebra gives (1.2), including all confluent cases. Finally the marked length-two incidence gives a finite birational resolution of the moving failure divisor. The power-basis determinant, confluent Vandermonde identity, and Fitting base change are classical algebraic tools. The assertions proved here concern their exact relative realization as multiplication failure schemes, their full primary fibres, and the normalization across contact collisions.

The later sections retain the quadratic component classification, its polar residual deformation theory and wall singularities, and the all-degree hyperplane theorem. The new conductor argument does not depend on the statistical interpretation or on those later classification results. Conversely, it does not identify the contact-subseries locus with the full ambient Grassmannian. The unrestricted quadratic excess scheme and the embedded primary ideal of the hyperplane family are different objects. Their previously proved statements are retained with their exact hypotheses. Section 15 distinguishes the classical inputs from the family-specific conclusions and records the remaining source-access limitation in the comparison with [4]. Full contact, realization and statistical applications are reproduced in the appendices.

2. EXACT CONDUCTOR REDUCTION IN FAMILIES

We work over $\mathbb{C}$. For a divisor $D \subset \mathbb{P}^1$ of length d write $I_D = \mathcal{O}(-D)$. The notation U^m means the image of $\text{Sym}^m U$ under multiplication, not an abstract symmetric power. For a finite scheme D , a subspace of $H^0(\mathcal{O}_D(n))$ is called generating if its evaluation map onto $\mathcal{O}_D(n)$ is surjective. This condition is open in a family. In relative notation, π denotes the projection from $\mathbb{P}^1$ times the parameter space; all bundles are pulled back to the indicated relative Grassmannian when needed.

Lemma 2.1 (Pencil multiplication). *If $U \subset V_n$ is base-point-free, then $UV_t = V_{n+t}$ for $t \geq n - 1$.*

Proof. Choose $0 \neq f \in U$. For each of its finitely many zeros the sections of U vanishing there form a proper hyperplane. Choose $g \in U$ outside their union. The binary forms f, g are coprime. The kernel of $(A, B) \mapsto fA + gB$ on $V_t \oplus V_t$ consists of $(gh, -fh)$ with $h \in V_{t-n}$, where this space is zero for $t < n$. For $t = n - 1$ the map is between two $2n$ -dimensional spaces and is injective. For $t \geq n$ its rank is $2(t + 1) - (t - n + 1) = n + t + 1$. In either case its image is V_{n+t} . $\square$

Theorem 2.2 (Relative conductor reduction). *Suppose $n \geq 2d - 1$, $m \geq 2$, and $1 \leq k \leq d$. On $\text{Hilb}^d(\mathbb{P}^1) = \mathbb{P}^d$ let $\mathcal{D}$ be the universal divisor and $\mathcal{E}_j = \pi_* \mathcal{O}_{\mathcal{D}}(j)$, of rank d . Let $X_{d,k}^\circ$ be the open relative Grassmannian of generating k -planes $\mathcal{A} \subset \mathcal{E}_n$, and define*

$$\mathcal{W} = \pi_* I_{\mathcal{D}}(n), \quad \mathcal{U} = \ker(V_n \otimes \mathcal{O} \longrightarrow \mathcal{E}_n / \mathcal{A}).$$

Then $\mathcal{U}$ has rank $n + 1 - d + k$, and

$$\text{im}(\mathcal{W} \otimes \text{Sym}^{m-1} \mathcal{U} \longrightarrow V_{mn} \otimes \mathcal{O}) = \pi_* I_{\mathcal{D}}(mn). \quad (2.1)$$

Restriction induces an isomorphism

$$\text{coker}(\text{Sym}^m \mathcal{U} \longrightarrow V_{mn} \otimes \mathcal{O}) \xrightarrow{\sim} \text{coker}(\text{Sym}^m \mathcal{A} \longrightarrow \mathcal{E}_{mn}). \quad (2.2)$$

Both assertions remain true after arbitrary base change to $X_{d,k}^\circ$. In particular the two cokernels have identical Fitting ideals in every index, including on nonreduced bases.

Proof. We first prove (2.1) on a geometric fibre. Write $D = (g)$, with g a binary form of degree d , so $W = gV_{n-d}$. The inverse-image subseries U is base-point-free: outside D this follows from W , and on D it follows from the generating hypothesis on A .

Products of two elements of W fill $g^2 V_{2n-2d}$. Since $2n - 2d \geq n - 1$, repeated applications of Lemma 2.1 give

$$W^2 U^{m-2} = g^2 V_{mn-2d} = H^0(I_D^2(mn)). \quad (2.3)$$

For $m = 2$ no application of the lemma is needed. For each further factor the input degree is $2n - 2d + jn \geq n - 1$.

The restriction of W to the first normal layer is surjective:

$$H^0(I_D(n)) \longrightarrow H^0((I_D/I_D^2)(n)).$$

Indeed the obstruction is $H^1(I_D^2(n)) = H^1(\mathcal{O}(n - 2d))$, which is zero because $n - 2d \geq -1$. A generating subspace A contains a section nonzero at all support points of D : avoid the finitely many evaluation hyperplanes. Such a section is a generator of the invertible $\mathcal{O}_D$ -module $\mathcal{O}_D(n)$. Lift it to $u \in U$. Multiplication by u^{m-1} identifies the first normal layer in degree n with that in degree mn . Consequently Wu^{m-1} surjects onto $H^0((I_D/I_D^2)(mn))$. This quotient is also the quotient of $H^0(I_D(mn))$ by $H^0(I_D^2(mn))$, since $mn - 2d \geq -1$. Combining this fact with (2.3) proves (2.1) on every geometric fibre.

All the targets just used are vector bundles on the parameter space: the requisite first cohomology groups vanish uniformly. The map in (2.1) is therefore a surjection of vector bundles by the fibrewise criterion and Nakayama's lemma. It stays surjective under every base change. The exact sequence $0 \rightarrow \mathcal{W} \rightarrow \mathcal{U} \rightarrow \mathcal{A} \rightarrow 0$ is locally split; hence the kernel of $\text{Sym}^m \mathcal{U} \rightarrow \text{Sym}^m \mathcal{A}$ is the image of $\mathcal{W} \otimes \text{Sym}^{m-1} \mathcal{U}$. Restrict the multiplication diagram to $\mathcal{D}$ and divide both source and target by these kernels. The surjectivity just proved identifies the two cokernels, giving (2.2). This argument is compatible with base change. Presentation independence and base change for Fitting ideals [25, Lemma 15.8.4, Tag 07ZA] finish the proof. $\square$

Remark 2.3 (What the reduction preserves). The theorem identifies coherent modules, not only the ranks of their fibres. Thus it preserves the equations of every corank scheme and all primary structure of its zeroth Fitting scheme on the stated base. It does not assert a primary decomposition for an arbitrary finite-dimensional multiplication system. Such a decomposition is

proved below for $k = 2$. The inequality $n \geq 2d - 1$ is a sufficient uniform bound; no assertion that it is necessary is used.

3. CONTACT PENCILS: EQUATIONS, PRIMARY COMPONENTS, AND INFINITESIMAL LAYERS

Fix $d \geq 4$ and $n \geq 2d - 1$. In this section the divisor D is fixed, although it need not be reduced. Set $G_D^\circ = \text{Gr}(2, E_{n,D})^\circ$. It is a smooth irreducible variety of dimension $2d - 4$. The conductor construction embeds it as the open generating locus in the Grassmannian of subseries containing W_D . Let $\mathcal{J}_m$ be the zeroth Fitting ideal of $\text{coker}(\text{Sym}^m \mathcal{U} \rightarrow V_{mn})$ on this variety.

Trivialize $\mathcal{O}_D(n)$, and put $R = H^0(\mathcal{O}_D)$. There is a smooth surjective frame map

$$\mathcal{F}_D = \{(a, v) : a \in R^\times, v \in R \setminus \mathbb{C}\} \longrightarrow G_D^\circ, \quad (a, v) \longmapsto \langle a, av \rangle. \quad (3.1)$$

It is the open part of the frame bundle where the first basis vector is a unit. Its relative dimension is four. All calculations on $\mathcal{F}_D$ below are therefore exact faithfully flat local calculations for G_D° , not transverse slices inferred from pointwise ranks.

Lemma 3.1 (The power-basis presentation). *On $\mathcal{F}_D$ the multiplication cokernel is presented by*

$$K_m(v) = [1, v, v^2, \dots, v^m] : \mathcal{O}^{m+1} \longrightarrow R \otimes \mathcal{O}. \quad (3.2)$$

For $m \geq d - 1$, every later column is an $\mathcal{O}$ -linear combination of the first d columns, and

$$\mathcal{J}_m \mathcal{O}_{\mathcal{F}_D} = (\Delta_D(v)), \quad \Delta_D(v) = \det[1, v, \dots, v^{d-1}]. \quad (3.3)$$

For $2 \leq m < d - 1$, $\mathcal{J}_m = 0$.

Proof. By Theorem 2.2, it suffices to multiply sections in the pencil $\langle a, av \rangle$ on D . Its image is generated by $a^m v^j$ for $0 \leq j \leq m$. Multiplication by a^{-m} is an invertible target operation and gives (3.2). Multiplication by v is an endomorphism of the rank- d free module $R \otimes \mathcal{O}$. Its monic characteristic polynomial annihilates it over the coefficient ring, by the adjugate proof of Cayley–Hamilton. Thus v^d and every higher power lie in the span of the first d powers, with regular coefficients. Column operations reduce the presentation to $[K_{d-1}(v), 0]$. This proves (3.3), as an identity of ideals and not merely of radicals. If $m + 1 < d$, all d -minors are zero. Faithful flatness descends this assertion. $\square$

Write $D = \sum_{i=1}^s d_i p_i$, with distinct p_i . Choose local parameters z_i and the Chinese-remainder identification

$$R = \prod_{i=1}^s \mathbb{C}[z_i]/(z_i^{d_i}), \quad v_i = \lambda_i + \sum_{r=1}^{d_i-1} c_{ir} z_i^r.$$

All λ_i, c_{ir} are independent affine coordinates before removing the scalar locus. For $d_i \geq 2$ define T_i by failure of A to separate $2p_i$; for $i < j$ define E_{ij} by failure to separate $p_i + p_j$. On the frame space they have equations $c_{i1} = 0$ and $\lambda_j - \lambda_i = 0$, respectively.

Theorem 3.2 (Full fixed-contact classification). *For every $m \geq d - 1$, in the ordered jet basis $1, z_i, \dots, z_i^{d_i-1}$,*

$$\Delta_D(v) = \prod_{i:d_i \geq 2} c_{i1}^{(d_i)} \prod_{i < j} (\lambda_j - \lambda_i)^{d_i d_j}. \quad (3.4)$$

The complete primary decomposition on G_D° is

$$\mathcal{J}_m = \bigcap_{i:d_i \geq 2} \mathcal{I}_{T_i}^{(d_i)} \cap \bigcap_{i < j} \mathcal{I}_{E_{ij}}^{d_i d_j}. \quad (3.5)$$

All displayed divisors are nonempty, smooth, irreducible, and distinct. They are precisely the associated subvarieties; no embedded associated points occur. The failure divisor has dimension $2d - 5$ and divisor class $\binom{d}{2}$ times the restricted Pluecker class.

Every multiplication corank, for every $m \geq 2$, is determined as follows. Put $e_i = \text{ord}_{z_i}(v_i - \lambda_i)$, with $e_i = d_i$ for a constant v_i , and put $\ell_i = \lceil d_i/e_i \rceil$. For each distinct value λ among the λ_i , set $L_\lambda = \max_{i:\lambda_i=\lambda} \ell_i$, and let $s(v) = \sum_\lambda L_\lambda$. Then

$$\text{corank}(\text{Sym}^m U_A \longrightarrow V_{mn}) = d - \min\{m + 1, s(v)\}. \quad (3.6)$$

On the frame space every higher Fitting ideal is, exactly, the corresponding minor ideal of (3.2).

Proof. For a polynomial P of degree at most $d - 1$, Taylor expansion gives

$$P(v_i(z_i)) = \sum_{r=0}^{d_i-1} \frac{P^{(r)}(\lambda_i)}{r!} (v_i(z_i) - \lambda_i)^r \pmod{z_i^{d_i}}.$$

The linear change from the divided derivatives of P at λ_i to the coefficients of $P(v_i(z_i))$ is triangular, with diagonal $1, c_{i1}, \dots, c_{i1}^{d_i-1}$. Its determinant is $c_{i1}^{\binom{d_i}{2}}$, including when this coefficient is zero. The determinant of the divided-derivative evaluation map

$$P \longmapsto \left(\frac{P^{(r)}(\lambda_i)}{r!} \right)_{i, 0 \leq r < d_i}$$

in the basis $1, T, \dots, T^{d-1}$ is the confluent Vandermonde $\prod_{i < j} (\lambda_j - \lambda_i)^{d_i d_j}$. Here is an explicit normalization of that identity. Start with d distinct variables, cluster d_i of them near λ_i , and in each cluster replace successive evaluation rows by divided-difference rows. The ordinary Vandermonde determinant is divided by the product of the differences *within* each cluster. In the cluster limit the rows are the divided derivatives above, while the remaining cross-cluster factors tend to the displayed product. This proves the identity on a dense open set and hence as a polynomial identity, including all collisions. Combining the two determinants proves (3.4) without inverting any of its factors.

For any length-two $Z \leq D$, restriction $E_{n,D} \rightarrow E_{n,Z}$ is surjective. The locus in $\text{Gr}(2, E_{n,D})$ where its restriction to A has rank at most one is the corresponding irreducible Schubert divisor. Its rank-zero locus is excluded by generation on D . Its rank-one locus is smooth: choose a basis of A with one nonzero image, and vary the second basis vector in a complementary target direction; the determinant then has a nonzero linear derivative. For irreducibility one may also mark the image line in $E_{n,Z}$; the resulting incidence is a Grassmannian bundle over $\mathbb{P}(E_{n,Z})$, and projects birationally onto the rank-one locus. The generating open is nonempty for each divisor when $d \geq 4$. On frames this follows directly by choosing $a = 1$ and a nonscalar v with the single prescribed linear equality. The same coordinates show that the divisors are distinct.

On each affine frame chart the coordinate ring is a localization of a polynomial ring. The distinct linear factors in (3.4) generate distinct prime ideals there. Unique factorization gives $(\prod f_\alpha^{b_\alpha}) = \bigcap (f_\alpha^{b_\alpha})$; each ideal $(f_\alpha^{b_\alpha})$ is primary. These identities descend by faithful flatness to (3.5). Equivalently, at each point of the smooth base the local ring is regular, each displayed divisor is smooth, and its powers are primary. This also proves absence of embedded primes. The dimension follows since this is a nonzero Cartier divisor. Each component is a Pluecker hyperplane section, and

$$\sum_i \binom{d_i}{2} + \sum_{i < j} d_i d_j = \binom{d}{2},$$

which proves the class formula.

Finally, in $\mathbb{C}[z_i]/(z_i^{d_i})$ the element $v_i - \lambda_i$ has nilpotency index ℓ_i . Its minimal polynomial is $(T - \lambda_i)^{\ell_i}$. In the product algebra the minimal polynomial is therefore

$$P_v(T) = \prod_\lambda (T - \lambda)^{L_\lambda}. \quad (3.7)$$

No polynomial of smaller degree can annihilate v , and division by this monic polynomial shows that $1, v, \dots, v^{s(v)-1}$ form a basis of $\mathbb{C}[v]$. Consequently the rank of $K_m(v)$ is $\min\{m+1, s(v)\}$. The cokernel isomorphism of Theorem 2.2 gives (3.6) and the assertion about Fitting ideals. $\square$

Theorem 3.3 (Exact infinitesimal layers and singularities). *Let $m \geq d-1$. At a point of a smooth frame chart discard the nonvanishing factors of (3.4). Write the remaining distinct linear factors as f_α , with multiplicities b_α , and put*

$$F = \prod_{\alpha} f_{\alpha}^{b_{\alpha}}, \quad P = \prod_{\alpha} f_{\alpha}.$$

In the regular local ring B of that chart the failure ring is $B/(F)$ and its nilradical is $\mathcal{N} = (P)/(F)$. Its exact nilpotency index is $\max_{\alpha} b_{\alpha}$, with index one for a reduced local ring. For every $q \geq 1$, there is an isomorphism of B -modules, with indicated generator P^q ,

$$\mathcal{N}^q / \mathcal{N}^{q+1} \simeq B / \left(P, \prod_{\alpha} f_{\alpha}^{(b_{\alpha}-q)+} \right). \quad (3.8)$$

The completed local equation is the same weighted linear arrangement, with unused frame and higher-jet coordinates as smooth parameters. Its tangent cone is defined by the homogeneous product of the active linear factors with these same multiplicities. Its singular scheme on this chart has ideal $(F, \partial F)$, and its singular support is the union of the components with $b_{\alpha} \geq 2$ and the pairwise intersections of distinct components. These descriptions descend smoothly to G_D° .

Proof. The local ring in question is a localization of a polynomial ring in the unit coordinates of a and the independent jet coordinates of v . The factors f_{α} are prime, distinct, and vanish at the selected point. Unique factorization shows that $P^q \in (F)$ if and only if $q \geq b_{\alpha}$ for every α . This proves the nilpotency assertion, including at multiple intersections.

The homomorphism $B \rightarrow \mathcal{N}^q / \mathcal{N}^{q+1}$ sending 1 to the class of P^q is onto. Its kernel is

$$((P^{q+1}, F) : P^q) = (P) + (F : P^q) = \left(P, \prod_{\alpha} f_{\alpha}^{(b_{\alpha}-q)+} \right).$$

For the middle equality, if $rP^q = sP^{q+1} + tF$, subtract sP from r ; cancellation gives membership in $(F : P^q)$. The last colon is computed by prime-factor exponents. This proves (3.8). Globally the chosen generator carries the corresponding line-bundle trivialization; the displayed isomorphism is a local one and makes no claim of a preferred global generator.

At the chosen point every active factor is linear in centered coordinates. Every discarded factor is a unit, so the completed ideal is exactly the displayed product, not a truncation of it. The initial ideal is generated by the same homogeneous product, which proves the tangent-cone statement. The hypersurface cotangent presentation gives $(F, \partial F)$. At a point on exactly one component of multiplicity one the differential is a unit times $df_{\alpha} \neq 0$. At every point on a multiple component or on two distinct components it is zero. This proves the singular support assertion. Smooth faithfully flat pullback preserves these local statements; in particular the four frame parameters do not alter the nilpotency index or the singular-support criterion. $\square$

For illustration, length four already gives subseries of codimension two, in every V_n with $n \geq 7$. For every $m \geq 3$ their fixed-contact primary equations are as follows. Put $\delta_{ij} = \lambda_j - \lambda_i$; only first-jet coefficients of multiple points are displayed.

Multiplicities of D	Exact equation on frames	Largest local nilpotency index
$(1, 1, 1, 1)$	$\prod_{i < j} \delta_{ij}$	1
$(2, 1, 1)$	$c_{11} \delta_{12}^2 \delta_{13}^2 \delta_{23}$	2
$(2, 2)$	$c_{11} c_{21} \delta_{12}^4$	4
$(3, 1)$	$c_{11}^3 \delta_{12}^3$	3
(4)	c_{11}^6	6

The reduced arrangement need not have normal crossings: three coincident values give three braid hyperplanes in a two-dimensional normal space. The equation, rather than a normal-crossing label, records that singularity. Formula (3.6) also distinguishes points with the same support of failure but different higher-jet vanishing orders.

4. MOVING CONTACTS AND THE NORMALIZATION OF THE FAILURE DIVISOR

Let $X_d^\circ = X_{d,2}^\circ$ be the relative generating Grassmannian in Theorem 2.2. It is smooth and irreducible of dimension $3d - 4$. The fixed-divisor schemes of the preceding section are its scheme-theoretic fibres, not its irreducible components. For $m \geq d - 1$ denote its failure scheme by Y_m .

4.1. Components over every multiplicity stratum. For a partition $\mathbf{d} = (d_1, \dots, d_s)$ of d , let $B_{\mathbf{d}} \subset \text{Sym}^d \mathbb{P}^1$ be the locally closed stratum of divisors with exactly this multiplicity partition, and restrict X_d° to it. Let M be the set of distinct multiplicities, and let $r_e = \#\{i : d_i = e\}$ for $e \in M$.

Corollary 4.1 (Complete component descent on multiplicity strata). *For $m \geq d - 1$, the failure scheme over $B_{\mathbf{d}}$ has*

$$\#\{e \in M : e \geq 2\} + \binom{\#M}{2} + \#\{e \in M : r_e \geq 2\} \quad (4.1)$$

irreducible components, all of dimension $s + 2d - 5$, and no embedded associated points. A ramification component is indexed by each $e \geq 2$ in M and has primary multiplicity $\binom{e}{2}$. An identification component is indexed by each unordered pair of multiplicities e, f that occurs at distinct support points and has primary multiplicity ef . The ideal of the scheme is the intersection of these prime-divisor ideals raised to the stated powers. On the labelled etale cover its full local equations and nilradical layers are exactly those of Theorem 3.3.

Proof. The ordered configuration space of s distinct points of $\mathbb{P}^1$ is an integral finite etale cover of $B_{\mathbf{d}}$, with deck group $\prod_{e \in M} \mathfrak{S}_{r_e}$. The relative first-jet and two-point restriction maps are surjective over this cover. Consequently each labelled T_i and E_{ij} is an irreducible relative rank-one Schubert divisor. The local factorization (3.4) holds with the support coordinates free, so there are no further components. The deck group permutes T_i exactly according to the value d_i , and permutes E_{ij} exactly according to the unordered pair $\{d_i, d_j\}$. The images of the orbit unions are precisely the irreducible components downstairs: each is the image of an integral labelled component, and two such images agree exactly for components in the same orbit. This gives (4.1).

Each component downstairs is a prime Cartier divisor in the smooth ambient space. Pulling its reduced ideal back to the labelled cover gives the reduced product of its labelled branch equations. Their multiplicity is constant on an orbit, so the primary ideal identity in (3.5) descends faithfully flatly. This also proves absence of embedded points and gives the stated local layers. The base stratum has dimension s , giving the dimension formula. A descended component can have intersecting local branches; smoothness of each labelled divisor is not a claim that every descended component is smooth. $\square$

Theorem 4.2 (An integral failure divisor with smooth normalization). *For $d \geq 4$, $n \geq 2d - 1$, and $m \geq d - 1$, the schemes Y_m are all the same integral Cartier divisor Y of dimension $3d - 5$. In particular Y has no embedded associated points. Its normalization is*

$$\tilde{Y} = \{(D, A, Z) : (D, A) \in X_d^\circ, Z \leq D, \deg Z = 2, \text{rank}(A \rightarrow H^0(\mathcal{O}_Z(n))) = 1\}. \quad (4.2)$$

The forgetful map $\tilde{Y} \rightarrow Y$ is finite and birational, and $\tilde{Y}$ is smooth and irreducible.

More explicitly, on an affine support chart write $D = (g(t))$ with g monic of degree d , and on a frame chart put $v = b/a \in \mathbb{C}[t]/(g)$, represented by a polynomial of degree less than d . The failure equation is

$$\Delta(g, v) = \det[1, v, \dots, v^{d-1}]_{\mathbb{C}[t]/(g)} = 0. \quad (4.3)$$

The normalization on this chart is parametrized by

$$g = hq, \quad v = \lambda + hw, \quad (4.4)$$

where h, q are monic of degrees $2, d - 2$, respectively, $\lambda \in \mathbb{C}$, and $\deg w \leq d - 3$. The scalar-pencil locus is omitted; the unit a is a free smooth parameter.

Proof. Theorem 2.2 applies on the whole relative space. The proof of Lemma 3.1 applies over the polynomial coefficient ring of a varying monic g , not only on its fibres. Thus every Y_m has the identical ideal (4.3) on a smooth covering by frame charts. This determinant is not identically zero: $v = t$ makes its matrix the identity for every g .

The space of pairs $Z \leq D$ with lengths $2, d$ is $\text{Sym}^2 \mathbb{P}^1 \times \text{Sym}^{d-2} \mathbb{P}^1$; the map sends (Z, Q) to $D = Z + Q$. It is finite. Indeed it is proper, and a fixed effective divisor has only finitely many degree-two subdivisors, obtained by choosing multiplicities at its finitely many support points. Restriction $E_{n,D} \rightarrow E_{n,Z}$ is a surjection of vector bundles. The rank-at-most-one condition on a pencil is a relative Schubert divisor; its rank-zero part is excluded by the generating hypothesis. This makes (4.2) a closed subscheme of the finite base change to X_d° , so its projection is finite.

Smoothness and irreducibility can be checked explicitly. A divisor with finite support is contained in an affine coordinate chart. On that chart Z is defined by a unique monic quadratic h , and $Z \leq D$ means $g = hq$. Since a is a unit, rank one on Z is equivalent to $v \bmod h$ being a scalar λ . Division by the monic h gives uniquely $v = \lambda + hw$, with the stated degree bound. Thus the framed incidence is an open part of affine space with coordinates the coefficients of h, q, w , the scalar λ , and the unit a . This proves smoothness. Globally irreducibility also follows from the relative rank-one Schubert description over the integral pair-divisor base; each fibre is irreducible, and its generating open is nonempty. These charts describe the same incidence, so no extra component appears at infinity.

An element v fails to generate the length- d algebra $R = H^0(\mathcal{O}_D)$ precisely when it either takes the same value at two distinct support points or has zero first derivative at a multiple support point. This follows at once from the minimal polynomial (3.7): its degree is d if and only if the support values are distinct and each local $\ell_i = d_i$. These two failure mechanisms are precisely the existence of a length-two $Z \leq D$ on which v is constant. Therefore the image of the finite incidence is the whole support of (4.3). That support is irreducible.

To determine its multiplicity, work over divisors with distinct support, label their roots after an étale base change, and use evaluation coordinates for v . The fixed-fibre formula becomes a product of the distinct differences $\lambda_j - \lambda_i$. At a point with exactly one equal pair the determinant has a nonzero linear derivative. Hence its multiplicity along its unique irreducible support is one. Since the ambient space is smooth, a Cartier hypersurface has no embedded associated points; a generically reduced such hypersurface is reduced. One may see the last assertions locally from unique factorization in the regular local ring, or from the Cohen–Macaulay and $R_0 + S_1$ criterion [24, Tag 031R]. Thus Y is integral.

Over the dense open subset with reduced D and precisely one equal pair there is exactly one possible Z . The finite projection is therefore birational in characteristic zero. Its source is smooth and hence normal, so it is the normalization. The affine parametrization is exactly (4.4), as already proved. The dimension of the divisor is $3d - 5$. $\square$

Proposition 4.3 (Exact singular-support test across collisions). *At a closed point $(D, A) \in Y$, call a length-two $Z \leq D$ scalar if A has rank one on Z . The point is smooth on Y if and only if there is precisely one scalar Z and the following linear system has no nonzero solution. In the associated factorization (4.4), the unknown is $r(t)$ of degree less than two, and the system is*

$$h \mid qr, \quad wr \bmod h \text{ is constant.} \quad (4.5)$$

In particular, this criterion is effective at nonreduced D and is not merely the distinct-support pair-count criterion. The full singular scheme on a frame chart has ideal

$$(\Delta, \partial\Delta/\partial g_0, \dots, \partial\Delta/\partial g_{d-1}, \partial\Delta/\partial v_0, \dots, \partial\Delta/\partial v_{d-1}). \quad (4.6)$$

Proof. If the normalization has more than one point above (D, A) , the local ring of Y cannot be regular. Suppose there is just one. We first justify the differential criterion for a finite normalization $\nu : N \rightarrow Y$ with N smooth. A regular local ring is normal, so a smooth point of Y has an isomorphic normalization near it. Conversely let $A_0 \subset B_0$ be the local finite extension at a point with a unique preimage and common residue field $\mathbb{C}$. Injectivity of $d\nu$ means

$$\mathfrak{m}_{B_0}/(\mathfrak{m}_{B_0}^2 + \mathfrak{m}_{A_0}B_0) = 0.$$

Nakayama's lemma first gives $\mathfrak{m}_{B_0} = \mathfrak{m}_{A_0}B_0$. It then gives, for the finite A_0 -module B_0 , that 1 generates B_0 , since the quotient by $\mathfrak{m}_{A_0}$ is $\mathbb{C}$. The injective map $A_0 \rightarrow B_0$ is consequently surjective. Thus $A_0 = B_0$ is regular. Local finiteness here follows by localizing the finite morphism; uniqueness of the preimage makes the resulting semilocal ring local.

In the affine normalization coordinates, a tangent vector with $\dot{g} = \dot{v} = 0$ satisfies

$$0 = q\dot{h} + h\dot{q}, \quad 0 = \dot{\lambda} + w\dot{h} + h\dot{w}.$$

Set $r = \dot{h}$, so $\deg r < 2$ because h is monic. The first equation is soluble exactly when $h \mid qr$, and then determines $\dot{q} = -qr/h$. The second is soluble exactly when the remainder of wr modulo h is constant; it determines $\dot{\lambda}$ as the negative of that constant and determines $\dot{w}$ by division. The resulting degrees satisfy the required bounds. If $r = 0$ all four variations are zero. Thus the differential has nonzero kernel exactly when (4.5) has a nonzero solution. The unit frame parameter has zero variation in a kernel and creates no additional one. Smooth frame descent proves the criterion on Y . Finally (4.6) is the hypersurface Jacobian presentation in its smooth affine ambient space. $\square$

Remark 4.4 (A smooth total point with a triple primary fibre). Take $d = 4$, $g = t^3(t - 1)$, and $A = \langle 1, t^2 \rangle$ on $D = (g)$. The only scalar length-two subdivisor is $h = t^2$. In (4.4) one has $q = t(t - 1)$ and $w = 1$. For $r = r_0 + r_1t$, divisibility $h \mid qr$ forces $r_0 = 0$, while constancy of $wr \bmod h$ forces $r_1 = 0$. Hence the total divisor is smooth here. On the fixed-divisor fibre, the two support values are distinct and (3.4) is a unit times c_{11}^3 . Its exact local nilpotency index is three. Thus even a smooth point of the integral total divisor can have a nonreduced primary fibre.

Proposition 4.5 (Discriminants and specialization). *Let $\chi_v(T)$ be the characteristic polynomial of multiplication by v in $\mathbb{C}[t]/(g)$. With the usual monic polynomial normalization of discriminants,*

$$\text{disc}(\chi_v) = \text{disc}(g) \Delta(g, v)^2. \quad (4.7)$$

This is a polynomial identity on the entire coefficient space. The equation $\Delta = 0$, not $\text{disc}(\chi_v) = 0$, defines the failure divisor, including when g has multiple roots. Its fibre over each fixed D is exactly the primary scheme in (3.5).

Proof. When g has distinct roots $t_1, \dots, t_d$, the evaluation matrix changes the basis $1, t, \dots, t^{d-1}$ to point values. Therefore

$$\Delta(g, v) = \frac{\prod_{i < j} (v(t_j) - v(t_i))}{\prod_{i < j} (t_j - t_i)}.$$

Squaring gives (4.7), because $\chi_v(T) = \prod_i (T - v(t_i))$ on this open set. Both sides are polynomial in the coefficients, so the identity holds everywhere. For $v = t$ the determinant is one even when g is nonreduced; this also shows why the factor $\text{disc}(g)$ must not be included in the failure equation. The scheme-theoretic fibre statement follows from Fitting base change in Theorem 2.2, or by specializing the polynomial determinant and applying (3.4). $\square$

Proof of Theorem 1.1. The fixed-contact assertions are Theorems 3.2 and 3.3. The moving-contact assertions are Theorem 4.2 and Proposition 4.3. All four concern the original multiplication map, by the coherent cokernel isomorphism of Theorem 2.2. $\square$

5. THE QUADRATIC FAMILY AND THE POLAR RESIDUAL FRAMEWORK

Let $V_n = H^0(\mathbb{P}^1, \mathcal{O}(n))$ and $k = n + 1$. Multiplication of sections defines, on the product of Grassmannians $X_L = \text{Gr}(c, V_n)^L$, the bundle map

$$\mu : \bigoplus_{\nu=1}^L \text{Sym}^2 \mathcal{U}_\nu \longrightarrow V_{2n} \otimes \mathcal{O}_{X_L}. \quad (5.1)$$

There are no products between different factors. We study the scheme $\mathcal{D}_L$ of maximal minors of this map, rather than only the rank at a general tuple. Put

$$q = \binom{c+1}{2}, \quad e = Lq, \quad p = 2k - 1, \quad M = Lc(k - c), \quad a = e - p + 1, \quad b = Lc - 3,$$

and assume $e \geq p$. The competition between a and b reflects two geometrically different kinds of annihilator. A nondegenerate Hankel form leads to the expected determinantal codimension a ; a rank-two form leads to signed secant families of codimension b . This competition is measured by multiplication on a smaller residual space and governs the singularities of the failure scheme.

The reduced support $\Delta^{(L)} = |\mathcal{D}_L|$ is the set of tuples for which $\sum_\nu U_\nu^2 \neq V_{2n}$. The basic codimension formula holds in the entire range $4 \leq c < k$.

Theorem 5.1 (Reduced failure locus and independent contacts). *Suppose $4 \leq c < k$, $L \geq 1$, and $Lq \geq p$. Then*

$$\text{codim}_{X_L} \Delta^{(L)} = \min\{Lq - p + 1, Lc - 3\}. \quad (5.2)$$

There is an irreducible common-secant subvariety $\Sigma_L \subset \Delta^{(L)}$ of codimension $Lc - 3$. If $Lc - 3 \leq Lq - p + 1$, it is a maximal-dimensional irreducible component of $\Delta^{(L)}$. Consequently a general L -tuple is informative if and only if $Lq \geq p$.

Theorem 11.1 refines this formula to a component classification in all these dimensions. For $a \leq b$ it determines all components and proves reducedness and Cohen–Macaulayness; for $b < a$ it determines all maximal-dimensional components. Exactly five additional equality incidences occur outside the two principal rank families. In each, the plane contains the full radical and is maximal isotropic in the quotient. The orientation covers in Section 10 determine their irreducibility.

The principal deformation result is independent of this binary classification. Given any map $\beta : \text{Sym}^m V \rightarrow F$, the plane derivative at an annihilator defines a polar residual space Q and

a map $\tau : Q \rightarrow \ker \ell$. Theorem 6.1 expresses the tangent excess of the incidence, and of its corank-one image, as $\dim \ker \tau$. In degree two,

$$Q = \bigoplus_{\nu} \operatorname{Sym}^2(U_{\nu} \cap \operatorname{rad} H_{\ell}),$$

so the obstruction is a smaller multiplication problem. The formulation also applies to products of sections on arbitrary projective varieties and to higher symmetric powers.

This tangent calculation has a scheme-theoretic global form. Theorem 6.2 determines the full singular support in expected codimension: all points of original corank at least two are singular, and a corank-one point is singular exactly when τ is not injective. On every constant-rank polar stratum, the Fitting ideals of the cotangent sheaf are the indicated minors of τ . These identities include the higher tangent-excess strata and glue with exact Schur–Jacobian presentations at larger corank. Corollary 6.3 consequently handles all annihilator ranks of the quadratic binary problem, including support collisions. Proposition 6.5 gives the corresponding local criterion in excess codimension.

The distinction between restricted ideals and neighbourhoods is addressed by Theorem 7.2. At arbitrary original corank, the joint polar relations are the obstruction to a transverse generic matrix chart. Even when that obstruction is nonzero, the theorem gives an exact matrix of residual convergent equations, retaining all higher terms and all nested rank ideals. In corank one, Theorem 7.5 reduces these to $\delta = \dim \ker \tau$ equations in $M - a + \delta$ parameters and identifies their quadratic obstruction. In expected codimension they form a regular sequence. Proposition 7.7 constructs transverse examples in every symmetric degree; the cubic example in Section 7 also exhibits a nontransverse cuspidal failure germ. These are exact local statements, not a claim that the binary higher-product component problem follows from the two-point calculation.

At a rank-two annihilator with distinct support, its radical is gV_{n-2} for the vanishing quadratic g . Writing $U_{\nu} \cap \operatorname{rad} H_{\ell} = gK_{\nu}$, the wall $a = b$ is exactly

$$L \binom{c}{2} = 2k - 5,$$

the square threshold for the residual multiplication of the $(c-1)$ -planes K_{ν} . Its entire determinantal rank stratification is the pullback of the corresponding cotangent Fitting stratification on the original corank-one locus (Corollary 6.4). On a nonempty dense open subset of its integral divisor, the analytic equation is $uv = 0$ (Theorem 12.9). The normal-bundle factorization, relative determinant line and simultaneous nonemptiness needed for this equation are proved separately in Section 12.3. We also give all same-annihilator signed intersections over the split rank-two base and an explicit primary decomposition of the fixed-annihilator isotropy fibre. These describe different objects: fibre embedded structure is not being asserted for the global scheme.

For higher symmetric powers the general polar theorem remains applicable. Section 13 additionally computes the two-point annihilator sectors by algebraic dihedral descent. This explicit family is not identified with all components of the higher-product failure scheme. The arbitrary- β theorem, rather than such an identification, is the general extension of the quadratic residual mechanism.

A complete higher-product family is provided by Theorem 14.1. For all hyperplanes $U \subset V_n$, $n \geq 4$, and every $m \geq 2$, the entire reduced failure locus of $\operatorname{Sym}^m U \rightarrow V_{mn}$ is the same secant threefold. We determine every corank, including the jump to m on the evaluation curve. In degree at least three we also determine all associated points of the projective failure scheme: the secant generic point and the embedded evaluation-curve generic point. A lower bound $\lceil m/2 \rceil$ for the nilpotency index shows that the high-degree scheme structure is not recovered from this fixed reduced support. This supplies a global excess-codimension consequence in every higher degree, beyond merely inserting a multiplication map into the general polar formalism.

The geometry arises naturally in recovery of calibrated information metrics: second-order distance contacts give the moment functional on the within-contact products U_ν^2 . A short application is given in Section 16. The appendices give the exact-contact, native-realization, boundary-experiment and finite-precision arguments, with their distinct hypotheses. None is used as an assumption in the algebraic proofs. Classical inputs and the precise extent of the primary-source comparison are recorded in Section 15.

6. POLAR RESIDUAL SPACES AND GLOBAL SINGULARITY STRATA

The deformation mechanism does not require binary forms. Let V, F be finite-dimensional complex vector spaces, $\dim F = p$, and fix a linear map

$$\beta : \mathrm{Sym}^m V \longrightarrow F, \quad m \geq 2.$$

Choose integers $0 < c_\nu < \dim V$, put $X = \prod_\nu \mathrm{Gr}(c_\nu, V)$, and write $M = \dim X$. The tautological multiplication map, its source rank and its expected failure codimension are

$$\mu : \mathcal{E} = \bigoplus_\nu \mathrm{Sym}^m \mathcal{U}_\nu \longrightarrow F \otimes \mathcal{O}_X, \quad e = \mathrm{rank} \mathcal{E}, \quad a = e - p + 1.$$

Assume $e \geq p$, and let $D = V(I_p(\mu))$. No assumption on the codimension of D is made until explicitly stated. The incidence $\mathcal{I} \subset \mathbb{P}(F^*) \times X$ is the zero scheme of $\ell\mu = 0$. Write $D^{(1)}$ for the locus where μ has corank one.

At a point $z = ([\ell], U)$ of this incidence, choose a representative ℓ and define the plane derivative

$$\begin{aligned} B_z : T_U X &\longrightarrow \mathcal{E}_U^*, \\ B_z(A)(u_1 \cdots u_m) &= \sum_{i=1}^m \ell \beta(u_1 \cdots A_\nu u_i \cdots u_m), \quad u_i \in U_\nu. \end{aligned} \quad (6.1)$$

This is well defined for $A_\nu : U_\nu \rightarrow V/U_\nu$: changing a lift adds a product annihilated by ℓ . Define the *polar residual space* and its multiplication map by

$$Q_z = \ker(B_z^* : \mathcal{E}_U \longrightarrow T_U^* X), \quad \tau_z = \mu_U|_{Q_z} : Q_z \longrightarrow \ker \ell \subset F. \quad (6.2)$$

The kernel Q_z is independent of the chosen nonzero representative of ℓ . On a constant-rank stratum of B , these spaces are vector bundles and τ is a bundle map. Equivalently, before trivializing the tautological line $\mathcal{L} = \mathcal{O}_{\mathbb{P}(F^*)}(-1)$, the plane derivative has target $\mathcal{E}^* \otimes \mathcal{L}^*$.

Theorem 6.1 (Polar residual tangent sequence). *At every point $z = ([\ell], U) \in \mathcal{I}$ there is an exact sequence*

$$0 \longrightarrow \ker B_z \longrightarrow T_z \mathcal{I} \longrightarrow (\mu_U Q_z)^\perp / \mathbb{C} \ell \longrightarrow 0. \quad (6.3)$$

In particular

$$\dim T_z \mathcal{I} = M - a + \dim \ker \tau_z. \quad (6.4)$$

The projection $\mathcal{I} \rightarrow D$ is a scheme isomorphism over $D^{(1)}$, so the same formula computes $\dim T_U D$ there. For $m = 2$, set $H_\ell(v, w) = \ell \beta(vw)$ and $W_\nu = U_\nu \cap \mathrm{rad} H_\ell$. Then

$$Q_z = \bigoplus_\nu \mathrm{Sym}^2 W_\nu, \quad \tau_z((q_\nu)_\nu) = \sum_\nu \beta(q_\nu). \quad (6.5)$$

Thus the quadratic obstruction is an actual smaller multiplication problem for every symmetric bilinear system, not only for binary forms.

Proof. Differentiating the incidence equations gives $\dot{\ell}\mu_U + B_z(\dot{U}) = 0$. A vector in $\mathcal{E}_U^*$ belongs to $\mathrm{im} B_z$ exactly when it annihilates Q_z . The equation is therefore soluble in $\dot{U}$ exactly when $\dot{\ell}$

annihilates $\mu_U Q_z$. Its set of solutions is a torsor under $\ker B_z$. Taking ℓ modulo $\mathbb{C}\ell$ proves (6.3). If $h = \dim Q_z$, then $\text{rank } B_z = e - h$, so its middle dimension is

$$M - (e - h) + p - 1 - \text{rank } \tau_z = M - a + \dim \ker \tau_z.$$

On a chart with an invertible $(p - 1)$ -minor, invertible row and column operations put μ in the form $\text{diag}(I_{p-1}, S)$, where S is a single row. The incidence equations first solve the normalized left kernel vector uniquely and then impose $S = 0$. This proves the scheme isomorphism, not just a bijection on points.

For the quadratic assertion, the map $V/U_\nu \rightarrow U_\nu^*$ induced by H_ℓ has image $(U_\nu/W_\nu)^*$. Its transpose has kernel W_ν , and it descends through U_ν because U_ν is isotropic. Choose $U_\nu = W_\nu \oplus Z_\nu$. Varying vectors of W_ν in lifts dual to Z_ν prescribes every mixed symmetric entry; varying the vectors of Z_ν prescribes their symmetric block, using division by two. Consequently $\text{im } B_z$ is exactly the space of forms vanishing on every $\text{Sym}^2 W_\nu$. Taking annihilators gives (6.5). $\square$

We use $I_s(A)$ for the ideal of s -minors of a matrix, with $I_s = (1)$ for $s \leq 0$ and $I_s = (0)$ if the requested minors are too large. For a finite module with presentation $R^q \rightarrow R^N \rightarrow C \rightarrow 0$, $\text{Fitt}_i C$ is generated by the $(N - i)$ -minors of the presentation matrix. This definition is independent of the presentation. All statements about restrictions of these ideals below mean pullback of the ideals of D ; they do not mean the intrinsic singular scheme of a locally closed stratum.

Theorem 6.2 (Global singularities and residual Fitting strata). *Suppose D is nonempty and of pure expected codimension $a > 0$ in X , and put $d = M - a$. Then its full singular support is*

$$\text{Sing}(D)_{\text{red}} = \{U : \text{corank } \mu_U \geq 2\}_{\text{red}} \cup \{U \in D^{(1)} : \ker \tau_{([\ell_U], U)} \neq 0\}_{\text{red}}. \quad (6.6)$$

The union denotes the indicated subsets of D , with their reduced closed union; the second set is locally closed in D . On any locally closed constant-rank stratum $T \subset D^{(1)}$ of B , with $h = \text{rank } Q$, one has, for every integer $j \geq 1$,

$$\text{Fitt}_{d+j-1}(\Omega_{D/\mathbb{C}})\mathcal{O}_T = I_{h-j+1}(\tau). \quad (6.7)$$

In particular the restriction of the singular scheme $V(\text{Fitt}_d \Omega_{D/\mathbb{C}})$ is the residual noninjectivity scheme $V(I_h(\tau))$, including its determinantal scheme structure. Formula (6.7) detects all tangent-excess strata, including points outside a generic node locus and points whose annihilator has colliding support.

At a point of original corank $\rho \geq 1$, write $S = D_0 - CA^{-1}B_0$ for the Schur complement of an invertible $(p - \rho)$ -block of μ . Let f_α be all its ρ -minors and choose smooth local coordinates $x_1, \dots, x_M$ on X . On this chart the singular scheme has the exact ideal

$$I_\rho(S) + I_a((\partial f_\alpha / \partial x_i)_{\alpha, i}) \quad \text{in } \mathcal{O}_X. \quad (6.8)$$

The Schur–Jacobian presentations hold on open charts for every $\rho \geq 1$, so they cover the whole failure scheme and are compatible on overlaps. The residual-minor simplification is a pullback to the stated strata, not an assertion of a product neighbourhood transverse to those strata.

Proof. Over $D^{(1)}$ use the incidence isomorphism of Theorem 6.1. In local coordinates on $\mathbb{P}(F^*) \times X$, the differential of its e equations is the map from a free module of rank $M + p - 1$ to one of rank e obtained by adjoining $\ell\mu$ to B . Its transpose presents the cotangent module of the incidence. Restrict to T and split off an invertible $(e - h)$ -minor of B . Constant rank means that its remaining Schur block is identically zero on T , not just at a chosen point. After elementary operations the remaining nonzero block is the map from $F^*/\mathbb{C}\ell$ to Q^* induced by $\ell\mu$; it is τ^* , with the tautological-line twist if no representative of ℓ was chosen. Fitting ideals commute with base change. The minors defining Fitt_{d+j-1} have size

$$(M + p - 1) - (d + j - 1) = e - j + 1.$$

After removing the identity block their size is $h - j + 1$. This proves (6.7). Change of frames or of a representative of ℓ multiplies the matrices by invertible matrices and hence leaves these ideals unchanged.

Since the local dimension on D is d , the tangent formula gives regularity at a corank-one point exactly when τ is injective. Over $\mathbb{C}$, regularity is smoothness. At a point of corank $\rho \geq 2$, Schur elimination identifies the defining ideal with $I_\rho(S)$, where every entry of S vanishes at the point. Thus this ideal is contained in the square of the maximal ideal. All its first differentials vanish, and $T_U D = T_U X$ has dimension $M > d$. This proves (6.6). Finally the usual exact cotangent presentation for the quotient by $(f_\alpha)_\alpha$ gives (6.8) directly from the definition of Fitt_d . The presentation independence of Fitting ideals proves compatibility with every other Schur chart and with (6.7). $\square$

Corollary 6.3 (Every annihilator rank for quadratic binary subseries). *In the notation of Theorem 5.1, suppose $4 \leq c < k$, $e \geq p$, and $a \leq b$. Theorem 6.2 applies to $\mathcal{D}_L$. On its corank-one locus, put $W_\nu = U_\nu \cap \text{rad } H_{\ell_\nu}$ and $h = \sum_\nu \binom{\dim W_\nu + 1}{2}$. Then*

$$U \text{ is smooth} \iff \bigoplus_\nu \text{Sym}^2 W_\nu \longrightarrow V_{2n} \text{ is injective.} \quad (6.9)$$

On each fixed-radical-intersection-dimension stratum, the singular ideal and all higher tangent-excess ideals are the minors in (6.7) of this smaller multiplication map. No assumption on distinct support, maximal annihilator rank, or membership in just one signed component is required.

Proof. Theorem 5.1 and Proposition 9.6 give pure expected codimension. Equation (6.5) identifies the residual map. Constant dimensions of the intersections give vector bundles after stratification. The preceding theorem applies. $\square$

Corollary 6.4 (The entire residual wall stratification). *At $a = b$, on a signed rank-two stratum with original multiplication of corank one and $U_\nu \not\subset \text{rad } H_\ell$, the pullback of the singular scheme is exactly the residual determinant scheme of μ'_K . More generally its tangent-excess-at-least- j ideal is $I_{h-j+1}(\mu'_K)$, where $h = L\binom{c}{2} = 2k - 5$. This includes every residual corank, not only the open ordinary-double-crossing locus.*

Proof. Here $Q = \bigoplus \text{Sym}^2(gK_\nu)$, and multiplication factors as μ'_K followed by the injective multiplication by g^2 . The source and residual target both have dimension h . Apply (6.7); multiplication by g^2 is a subbundle inclusion on the split rank-two base. $\square$

Proposition 6.5 (Excess-codimension form of the residual criterion). *Without the expected-codimension hypothesis, let $T \subset D^{(1)}$ be a constant-rank stratum of B . For any integer t , locally*

$$\text{Fitt}_{M-t}(\Omega_{D/\mathbb{C}})\mathcal{O}_T = I_{h-a+t}(\tau). \quad (6.10)$$

At a point where $\dim_U D = M - t$, smoothness is equivalent to $\dim \ker \tau = a - t$. For quadratic binary subseries in the excess regime $b < a$, this gives $t = b$ at every corank-one point lying on a maximal-dimensional component. It is a statement about the full local failure scheme there, not only its selected component.

Proof. The same cotangent presentation has $M + p - 1$ generators, so the requested minors have size $p - 1 + t = e - a + t$. Remove the $(e - h)$ -identity block. This gives (6.10). Compare (6.4) with the stated local dimension. In the last assertion the global dimension bound is $M - b$, and the component through the point attains it. $\square$

These results separate a full singular-locus calculation from a classification of local analytic types or of associated primes. Equation (6.8) retains nonlinear Schur entries: replacing it by minors of their linear parts would lose information. Neither the global criterion nor the excess formula asserts that every germ is a normal crossing, or that every excess scheme is unmixed.

The generality in β has concrete instances. For a projective variety Y and a line bundle A , one may take $V = H^0(Y, A)$, $F = H^0(Y, A^m)$ and β to be multiplication of sections; the same construction applies to multivariate polynomial spaces and their linear quotients. Theorems 6.1 and 6.2 use precisely the hypotheses stated above in each case. They do not require a secant parameterization or a classification of catalecticant ranks. For $m > 2$ the polar residual Q need not be a symmetric power of a radical, which is why the general statement uses the kernel of the plane derivative instead of imposing a quadratic description.

7. INTRINSIC RESIDUAL EQUATIONS AT EVERY MULTIPLICATION CORANK

The residual Fitting identities describe restrictions of intrinsic ideals. To pass from these restrictions to a neighbourhood, one must retain the nonlinear equations. We give such a presentation at every point. The linear algebra in this section applies to a map of vector bundles on any smooth complex variety; the polar formula identifies its intrinsic obstruction for the multiplication systems of Section 6.

The matrix normal derivative and the transverse determinantal model used below are classical [10, Remark 1.18; Section 1.3.2]. The point here is their explicit expression in multiplication polars, including the joint relations between distinct annihilators, and the nonlinear equations left when transversality fails.

7.1. The joint polar obstruction. Fix $U \in D$, and write

$$K_U = \ker \mu_U, \quad C_U = \operatorname{coker} \mu_U, \quad \rho = \dim C_U, \quad s = \dim K_U = e - p + \rho.$$

Here $C_U^* \subset F^*$ is the entire space of annihilators, not a choice of one projective annihilator. Differentiating μ and then restricting to its kernel and projecting to its cokernel defines

$$\partial_U : T_U X \longrightarrow \operatorname{Hom}(K_U, C_U), \quad A \longmapsto [\dot{\mu}_U(A)|_{K_U}]. \quad (7.1)$$

This map is intrinsic. Indeed a change of source or target frame changes $\dot{\mu}$ by terms of the form $P\mu_U + \mu_U Q$; these vanish after restriction to K_U and projection to C_U .

Proposition 7.1 (Joint polar relations). *For a symmetric multiplication system, the transpose of (7.1) is*

$$\begin{aligned} \partial_U^* : K_U \otimes C_U^* &\longrightarrow T_U^* X, \\ \sum_i q_i \otimes \ell_i &\longmapsto \sum_i B_{([\ell_i], U)}^*(q_i), \end{aligned} \quad (7.2)$$

where the representative ℓ_i is used in the definition of B ; the expression is linear in that representative and extends to $\ell_i = 0$. Put

$$\mathcal{R}_U = \ker \partial_U^*, \quad t = \operatorname{rank} \partial_U, \quad b_U = \dim \mathcal{R}_U = \rho s - t.$$

If $\rho = 1$, a choice of nonzero annihilator identifies $\mathcal{R}_U$ with $\ker \tau_{([\ell], U)}$. If $\rho > 1$, the kernel in (7.2) includes relations between distinct annihilators; it is not in general the direct sum of the single-annihilator kernels.

Proof. For $q \in K_U$, $\ell \in C_U^*$ and $A \in T_U X$, the natural trace pairing gives

$$\langle q \otimes \ell, \partial_U(A) \rangle = \ell \dot{\mu}_U(A)(q) = B_{([\ell], U)}(A)(q).$$

Summing proves the formula. If $C_U^* = \mathbb{C}\ell$, its kernel is $K_U \cap \ker B^* = K_U \cap Q = \ker(\mu_U|_Q)$. All identifications are compatible with changes of frame by the intrinsic construction of ∂_U . $\square$

7.2. An exact residual presentation. Let $H_U = \text{Hom}(K_U, C_U)$ and $H_0 = \text{im } \partial_U$. Choose a vector-space complement H_1 so that $H_U = H_0 \oplus H_1$, and choose an isomorphism $L : \mathbb{C}^t \rightarrow H_0$. Such choices specify coordinates, not additional hypotheses. In particular H_1 identifies with coker ∂_U and has dimension b_U .

Theorem 7.2 (Residual matrix germ). *At every $U \in D$ there are analytic coordinates $(y_1, \dots, y_t, x_1, \dots, x_{M-t})$ centred at U and an H_1 -valued convergent power series $\Psi(y, x)$ with $\Psi(0) = d\Psi(0) = 0$ such that*

$$\widehat{\mathcal{O}}_{D,U} \simeq \mathbb{C}[[y_1, \dots, y_t, x_1, \dots, x_{M-t}]]/I_\rho(L(y) + \Psi(y, x)). \quad (7.3)$$

The same presentation holds with convergent power series for the analytic local ring. Moreover, for $1 \leq j \leq \rho$, the ideal of the locus where the original multiplication has rank at most $p - \rho + j - 1$ is, in this presentation,

$$I_j(L(y) + \Psi(y, x)). \quad (7.4)$$

These are equalities of ideals after analytic change of coordinates, not merely tangent-space or set-theoretic assertions. No expected-codimension assumption is required.

Proof. Put $r = p - \rho$. Trivialize source and target and choose an invertible r -minor at U . After constant changes of frame the matrix is

$$\mu = \begin{pmatrix} A & B_0 \\ C & D_0 \end{pmatrix}, \quad A(U) = I_r, \quad B_0(U) = C(U) = D_0(U) = 0.$$

Invertible analytic row and column operations reduce it to $\text{diag}(I_r, S)$, where $S = D_0 - CA^{-1}B_0$ is a $\rho \times s$ matrix vanishing at U . If $r = 0$, this means $S = \mu$, with no inverse block. The differential of S at U is precisely (7.1); the correction $CA^{-1}B_0$ starts in degree two.

Let $P_0 : H_U \rightarrow H_0$ and $P_1 : H_U \rightarrow H_1$ be the projections. The map $L^{-1}P_0S$ has differential of rank t . Complete its t coordinate functions to analytic local coordinates (y, x) on X using the inverse function theorem. Then $P_0S = L(y)$ exactly, whereas $\Psi = P_1S$ has no constant or linear term. This proves the claimed matrix expression. For a block diagonal matrix with an invertible r -block,

$$I_{r+j}(\text{diag}(I_r, S)) = I_j(S) \quad (1 \leq j \leq \rho).$$

To see both inclusions, a j -minor of S extends by the identity block to an $(r+j)$ -minor; all other terms are minors of S of size at least j and belong to $I_j(S)$. This proves (7.4) and, with $j = \rho$, the analytic presentation of D . Completion gives (7.3). $\square$

Remark 7.3. The power series Ψ is not a canonical function: complements, frames and coordinates enter its construction. The completed local ring and the nested determinantal ideals are canonical. Their compatibility on overlaps follows because each presentation is obtained from the original ideals $I_{r+j}(\mu)$ by invertible operations. The theorem therefore supplies local equations also at support collisions and at points with several annihilators. It does not replace Ψ by its quadratic term or assert that a constant polar-rank stratum has a product neighbourhood.

The first nonzero residual terms can be computed without solving all the equations. For example, along a curve with parameter h , write

$$\mu(h) = \begin{pmatrix} A_0 + hA_1 + O(h^2) & hB_1 + h^2B_2 + O(h^3) \\ hC_1 + h^2C_2 + O(h^3) & hD_1 + h^2D_2 + O(h^3) \end{pmatrix}, \quad A_0 \text{ invertible.}$$

Expansion of the inverse gives

$$S(h) = hD_1 + h^2(D_2 - C_1A_0^{-1}B_1) + O(h^3). \quad (7.5)$$

For a velocity in $\ker \partial_U$, the class of the second coefficient in H_U/H_0 is independent of the acceleration of the curve. An acceleration changes that coefficient by an element of $\text{im } \partial_U$. The

same argument, using the chain rule and $D_1 = 0$, proves independence of coordinates and frames. Thus (7.5), paired with $\mathcal{R}_U = (H_U/H_0)^*$, is the intrinsic quadratic obstruction on $\ker \partial_U$. It describes the quadratic part of $\Psi(0, x)$; it is not by itself the initial ideal of all the maximal minors.

Corollary 7.4 (Transverse determinantal neighbourhoods). *If $\mathcal{R}_U = 0$, then $M \geq \rho s$ and*

$$\widehat{\mathcal{O}}_{D,U} \simeq \mathbb{C}[[z_1, \dots, z_{M-\rho s}, (y_{ij})_{1 \leq i \leq \rho, 1 \leq j \leq s}]]/I_\rho((y_{ij})). \quad (7.6)$$

This is also an analytic product decomposition. The germ is reduced, Cohen–Macaulay and of dimension $M - a$. If $\rho = 1$ it is smooth; if $\rho \geq 2$ its singular support is the product of the smooth factor with the rank-at-most- $(\rho - 2)$ generic matrix locus. This last assertion concerns reduced support, not equality of the singular-scheme ideal with $I_{\rho-1}((y_{ij}))$.

Proof. Here $t = \rho s$, $H_1 = 0$, and (7.3) has no nonlinear residual part. Choosing the entries of S as coordinates proves (7.6). For completeness, the generic rank-at-most- $(\rho - 1)$ matrix variety is irreducible: it is the closure of products of a $\rho \times (\rho - 1)$ matrix and a $(\rho - 1) \times s$ matrix. Its rank- $(\rho - 1)$ stratum has dimension $(\rho - 1)(s + 1)$, as one sees by choosing its image and a surjection onto it. Its codimension is $s - \rho + 1 = a$. The expected-grade maximal-minor resolution [8] consequently gives a Cohen–Macaulay quotient. At rank $\rho - 1$, Schur elimination leaves a independent entries, so the quotient is generically smooth and hence generically reduced. Cohen–Macaulayness gives S_1 ; $R_0 + S_1$ implies reducedness [24]. At rank at most $\rho - 2$, every first derivative of every maximal minor vanishes. The tangent dimension is the ambient dimension, strictly larger than the local dimension. These and only these points are singular. For $\rho = 1$ the ideal consists of independent coordinates. The dimension and all product assertions follow. $\square$

7.3. Corank-one equations and their quadratic obstruction. Corank one admits a smaller presentation than the general matrix normal form: all independent linear equations may be eliminated.

Theorem 7.5 (Minimal residual equations in expected codimension). *Suppose μ_U has corank one, put $z = ([\ell], U)$ and $\delta = \dim \ker \tau_z$, and let $d = M - a$. There is an exact presentation*

$$\widehat{\mathcal{O}}_{D,U} \simeq \mathbb{C}[[x_1, \dots, x_{d+\delta}]]/(f_1, \dots, f_\delta), \quad f_i \in (x_1, \dots, x_{d+\delta})^2, \quad (7.7)$$

and an analogous convergent presentation. If D has codimension a at U , these δ equations form a regular sequence. In that case the displayed embedding dimension and number of equations are minimal. Without that codimension hypothesis the exact presentation still holds, but regularity of the sequence is not asserted.

The quadratic obstruction takes values in $(\ker \tau_z)^$. For a tangent vector $(\lambda, A) \in T_z \mathcal{I}$, choose graph lifts of the maps $A_\nu : U_\nu \rightarrow V/U_\nu$. On a monomial $u_1 \cdots u_m \in \text{Sym}^m U_\nu$ define*

$$\begin{aligned} \mathcal{H}_{\lambda,A}(u_1 \cdots u_m) &= \sum_{i < j} \ell \beta(u_1 \cdots A_\nu u_i \cdots A_\nu u_j \cdots u_m) \\ &\quad + \sum_i \lambda \beta(u_1 \cdots A_\nu u_i \cdots u_m). \end{aligned} \quad (7.8)$$

Extend linearly to $\mathcal{E}_U$. Its restriction to $\ker \tau_z$ is the intrinsic quadratic obstruction and equals the degree-two part of the residual equations, after identifying their coordinate space with $(\ker \tau_z)^$.*

Proof. The incidence is the zero scheme of a rank- e section on the smooth $(M + p - 1)$ -dimensional variety $\mathbb{P}(F^*) \times X$. Its differential is

$$T_{[\ell]} \mathbb{P}(F^*) \oplus T_U X \longrightarrow \mathcal{E}_U^*, \quad (\lambda, A) \longmapsto \lambda \mu_U + B_z(A).$$

The annihilator of its image is $\ker \mu_U \cap \ker B_z^* = \ker \tau_z$; thus its rank is $e - \delta$. Choose that many independent equations and eliminate them, together with the corresponding coordinates, by the

analytic implicit function theorem. The resulting smooth space has dimension $M + p - 1 - (e - \delta) = d + \delta$. Subtracting appropriate linear combinations of the eliminated equations leaves δ equations with zero differential. Projection from the incidence is a scheme isomorphism here by Theorem 6.1; this proves (7.7).

If the local codimension is a , this local ring has dimension d . In the regular local ring with $d + \delta$ parameters its ideal, generated by δ elements, has height δ . Such a sequence is regular: a regular local ring is Cohen–Macaulay, and a height-equal-to-length generating sequence is a partial system of parameters, hence a regular sequence. Since all f_i are quadratic or higher, no further smooth coordinate can be eliminated, and a regular sequence of this length minimally generates the ideal.

To compute the quadratic terms, trivialize the projective line near $[\ell]$ and use the graph path $(\ell + h\lambda, u_i + hA_\nu u_i)$. Expanding the incidence section, its coefficient of h^2 is exactly (7.8). The order- h coefficient vanishes because (λ, A) is tangent. A second-order change of path changes the coefficient of h^2 by the image of the section's differential, which is annihilated by $\ker \tau_z$. Changes of graph lifts, projective trivialization or frames have precisely the same effect, by the chain rule and the vanishing first-order section. Thus the restriction is well defined. The nonlinear corrections used in eliminating the independent equations are again second-order changes of path and have zero pairing with this kernel. This proves the asserted identification with the degree-two residual equations. $\square$

Remark 7.6. Theorem 7.5 identifies an actual complete-intersection germ in expected codimension, not just its tangent dimension. Its quadratic forms need not generate the full initial ideal. Higher-order terms in the f_i or in Ψ must be retained unless an additional normal-form argument removes them. The simple-normal-determinant argument below supplies such an argument on the residual wall. In the excess regime the same exact equations remain available without silently imposing expected grade.

7.4. Realization in every symmetric degree. The transverse class is not confined to two-point annihilators or to quadratic binary multiplication. The following explicit model realizes it for every symmetric degree and every possible multiplication corank.

Proposition 7.7 (Realization of transverse matrix germs). *Let $m \geq 2$, $e \geq p \geq 1$ and $1 \leq \rho \leq p$ be integers. There exist a vector space V , a surjection $\beta : \text{Sym}^m V \rightarrow \mathbb{C}^p$, and e one-dimensional contacts in $X = \text{Gr}(1, V)^e$ for which the multiplication has corank ρ and $\mathcal{R}_U = 0$. Consequently its failure germ is the full generic $\rho \times (e - p + \rho)$ maximal-minor germ, times a smooth factor, for every m .*

Proof. Put $r = p - \rho$ and take V with basis $x_1, \dots, x_e, w_1, \dots, w_\rho$. Let $f_1, \dots, f_p$ be the target basis. Define β on monomials by

$$\begin{aligned} \beta(x_i^m) &= f_i & (1 \leq i \leq r), \\ \beta(x_j^{m-1} w_\alpha) &= \frac{1}{m} f_{r+\alpha} & (r < j \leq e, 1 \leq \alpha \leq \rho), \end{aligned} \tag{7.9}$$

and set it equal to zero on all remaining basis monomials. The listed monomials are distinct, including when $m = 2$, so this defines a linear map. It is surjective because $e - r = e - p + \rho \geq \rho > 0$. At $U_j = \mathbb{C}x_j$, the source of the contact multiplication has basis $(x_j^m)_j$, its first r columns are $f_1, \dots, f_r$, and its remaining columns are zero. Thus K_U is spanned by the last $s = e - r$ source vectors and C_U by $f_{r+1}, \dots, f_p$. Varying only the j th line from x_j toward w_α has derivative

$$\left. \frac{d}{dh} \right|_{h=0} \beta((x_j + hw_\alpha)^m) = f_{r+\alpha}, \quad j > r.$$

These ρ s independent tangent directions map under ∂_U to every matrix unit in $\text{Hom}(K_U, C_U)$. The map is surjective, and Corollary 7.4 applies. $\square$

This proposition concerns arbitrary symmetric linear systems. It exhibits genuine higher-degree failure germs and supplies concrete models for the intrinsic theorem. It is separate from, and does not presuppose, a global classification for $\text{Sym}^m U \rightarrow V_{mn}$ on the binary Grassmannian. The latter system's two-point sectors remain the subject of Section 13.

7.5. Two explicit nontransverse residual germs. These familiar singularities provide direct tests of the distinction between a residual space and the nonlinear residual equations.

A simultaneous-annihilator relation. For the bundle matrix on $\mathbb{A}^3$

$$\mu = \begin{pmatrix} x & y \\ y & z \end{pmatrix},$$

the origin has $K = C = \mathbb{C}^2$. The intrinsic derivative is inclusion of the symmetric matrices into all 2×2 matrices. Its transpose has the one-dimensional kernel generated by $e_1 \otimes e_2^* - e_2 \otimes e_1^*$. For each fixed nonzero $\ell \in C^*$, however, the map $K \rightarrow T_0^* \mathbb{A}^3$, $q \mapsto \partial^*(q \otimes \ell)$, is injective: its coefficients are those of the product of two nonzero linear binary forms. Thus checking one annihilator at a time would miss the joint relation. In Theorem 7.2, H_0 is the symmetric subspace and $\Psi = 0$; the exact failure germ is $xz - y^2 = 0$. This example is a bundle-matrix example for the intrinsic theorem, not an additional claim about binary subseries.

A cubic multiplication cusp. Let $V = \mathbb{C}X \oplus \mathbb{C}Y \oplus \mathbb{C}Z$, $F = \mathbb{C}$, and define $\beta : \text{Sym}^3 V \rightarrow F$ by $\beta(Y^2Z) = 1/3$, $\beta(X^3) = -1$, and zero on the other basis monomials. Take one line contact, and use the chart $U = \mathbb{C}(Z + xX + yY)$ near $\mathbb{C}Z$. Its multiplication matrix is the one-entry matrix

$$\beta((Z + xX + yY)^3) = y^2 - x^3.$$

At the origin $\delta = 1$ and the quadratic obstruction is y^2 , whereas the exact residual equation is $y^2 - x^3$. The map from its completed local ring to $\mathbb{C}[[t]]$ given by $x = t^2$, $y = t^3$ is injective: reducing a power series modulo $y^2 - x^3$ leaves $A(x) + yB(x)$, and the even powers in $A(t^2)$ cannot cancel the odd powers in $t^3B(t^2)$. Thus the cusp has one branch, unlike the double crossing. This explicit higher-product system shows why the full residual equation, rather than only its quadratic part, occurs in the theorem.

8. THE GLOBAL MULTIPLICATION FAILURE LOCUS

Throughout this section the field is $\mathbb{C}$, $k = n + 1$, $4 \leq c < k$, $q = \binom{c+1}{2}$, and $p = 2k - 1$. On $X_L = \text{Gr}(c, V_n)^L$ let $\mathcal{S}_\nu$ denote the tautological bundles. The failure scheme is defined by the maximal minors of

$$\mu : \bigoplus_{\nu=1}^L \text{Sym}^2 \mathcal{S}_\nu \longrightarrow V_{2n} \otimes \mathcal{O}_{X_L}.$$

Its underlying set is $\Delta^{(L)}$. We shall prove Theorem 5.1 without using a maximal-rank theorem for general curves.

8.1. Annihilating Hankel forms. For $\ell \in V_{2n}^*$, let $H_\ell(f, g) = \ell(fg)$ on V_n . The following classical input concerns the whole rank locus, including confluent and limiting forms; it is not an assertion only about sums of distinct point evaluations.

Lemma 8.1 (Classical Hankel rank strata). *In $\mathbb{P}(V_{2n}^*)$, the rank-exactly- r stratum of H_ℓ is nonempty and has dimension*

$$h_r = \begin{cases} 2r - 1, & 1 \leq r < k, \\ 2k - 2, & r = k. \end{cases}$$

For $r < k$, its closure is the variety of sums of at most r point evaluations, with Zariski closure understood.

Identification with the classical statement. The rank- $\leq r$ equations are the $(r+1)$ -minors of the $k \times k$ Hankel matrix in $2k-1$ independent coordinates. The equality of Hankel minor ideals under changing the rectangular shape, recalled in [7, Section 1], identifies their coordinate ring with that of the maximal minors of an $(r+1) \times (2k-r-1)$ Hankel matrix. That ring has affine dimension $2r$. The secant identification is [7, Section 2, equation (2.0.1)]; thus the projective dimension is $2r-1$. The rank- $\leq r-1$ subvariety has smaller dimension, so the exact-rank stratum is nonempty and dense. The full-rank stratum is open in $\mathbb{P}^{2k-2}$. These are classical Hankel determinantal facts; no claim about their rational singularities is needed here. $\square$

Lemma 8.2 (Isotropic incidence dimensions). *Let a symmetric form on a k -space have rank r and radical E of dimension $k-r$. For a totally isotropic c -plane U , put*

$$s = \dim((U + E)/E), \quad a = c - s.$$

Such planes exist precisely for

$$\max\{0, c - k + r\} \leq s \leq \min\{c, \lfloor r/2 \rfloor\}. \quad (8.1)$$

The stratum with this value of s has codimension

$$D_{r,s} = (c-s)(r-s) + \frac{s(s+1)}{2} \quad (8.2)$$

in $\text{Gr}(c, k)$.

Proof. Choose $U \cap E$, an a -plane in E , with dimension $a(k-r-a)$. The image of U in the nondegenerate r -dimensional quotient is an isotropic s -plane, whose Grassmannian has dimension $s(r-s) - s(s+1)/2$. This follows by differentiating the symmetric restriction equations at an isotropic frame: pairing with a complementary dual frame gives all symmetric variations, so the equations are independent. Existence is equivalent to $2s \leq r$. Finally the lifts are an affine space of dimension $s(k-r-a)$. Subtracting the sum from $c(k-c)$ gives (8.2). Maximal orthogonal Grassmannians may have two components; both have this same dimension, which is all the count uses. $\square$

Lemma 8.3 (Rank optimization). *Assume $Lq \geq 2k-1$. For all r, s satisfying (8.1),*

$$LD_{r,s} - h_r \geq \min\{Lq - 2k + 2, Lc - 3\}. \quad (8.3)$$

Proof. Write $A_0 = Lq - 2k + 2$ and $B_0 = Lc - 3$. If $r = k$, feasibility forces $s = c$, giving exactly A_0 . Suppose $r < k$. For $s = 0$, the left side is $(Lc - 2)r + 1 \geq Lc - 1 > B_0$.

For $1 \leq s \leq c-2$, the coefficient of r in $LD_{r,s} - 2r + 1$ is $L(c-s) - 2 \geq 0$. Since $r \geq 2s$, the expression is at least

$$g(s) = 1 + s(Lc - 4) - \frac{Ls(s-1)}{2}.$$

This is concave as a function of s , and

$$g(1) = B_0, \quad g(c-2) - B_0 = (c-3) \left(\frac{L(c+2)}{2} - 4 \right).$$

The latter is nonnegative except when $(L, c) = (1, 4)$ or $(1, 5)$. In the first exception, saturation forces $k = 5$ and feasibility forces $s \leq 1$. In the second, $k \leq 8$; the values $g(1), g(2), g(3)$ are 2, 2, 1. The last value is feasible only when $k = 8$, in which case $A_0 = 1$. Both exceptions therefore satisfy (8.3).

For $s = c - 1$ and $L = 1$, the expression decreases with r , hence is at least its value at $r = k - 1$, namely $A_0 + k - 2c + 1$. Feasibility implies $k \geq 2c - 1$. For $s = c - 1$ and $L \geq 2$, it is minimized at $r = 2c - 2$ and exceeds B_0 by

$$(c - 2) \left(\frac{L(c + 1)}{2} - 4 \right) \geq 0.$$

Finally, if $s = c$ and $r < k$, it is $Lq - 2r + 1 \geq A_0 + 1$. These cases exhaust (8.1). $\square$

8.2. The common-secant family. Let $C_n \subset \mathbb{P}(V_n^*)$ be the evaluation rational normal curve, and let S_n be the closure of its secant lines. Since $n \geq 4$, S_n is irreducible of dimension three. Define

$$\mathfrak{S}_L = \{([h], U_1, \dots, U_L) : [h] \in S_n, U_\nu \subset \ker h\}, \quad \Sigma_L = \text{pr}_{X_L} \mathfrak{S}_L. \quad (8.4)$$

Proposition 8.4 (A global secant obstruction). *The closed set Σ_L is irreducible, has codimension $Lc - 3$ in X_L , and lies in $\Delta^{(L)}$.*

Proof. The incidence is a relative product of Grassmannians over S_n , so it is irreducible of dimension $3 + Lc(k - 1 - c)$ and its projection is closed. For a general $h = \alpha \text{ev}_x + \beta \text{ev}_y$, with $x \neq y$ and $\alpha\beta \neq 0$, every $f \in \ker h$ satisfies $\alpha f(x) = -\beta f(y)$. Consequently

$$\ell = \alpha^2 \text{ev}_x^{(2n)} - \beta^2 \text{ev}_y^{(2n)}$$

is nonzero and annihilates all products from each U_ν . Closedness extends this conclusion to the whole incidence.

It remains to check the dimension of the image, rather than merely that of the incidence. Fix a general h . If $c = k - 1$, the hyperplane U_1 determines h . Otherwise, the c -planes in $\ker h$ that are also annihilated by some $h' \in S_n \setminus \{h\}$ form a constructible set of dimension at most

$$3 + c(k - 2 - c) < c(k - 1 - c),$$

because $c \geq 4$. Here, for each h' , the plane lies in the intersection of two distinct hyperplanes. Thus a general $U_1 \subset \ker h$ admits no such h' , and the incidence projection is generically one-to-one. Subtraction from $\dim X_L = Lc(k - c)$ gives $Lc - 3$. $\square$

Proof of Theorem 5.1. Failure is equivalent to the existence of a nonzero ℓ annihilating every U_ν^2 . Consider the closed projective incidence of such pairs $((U_\nu), [\ell])$. On the rank- r stratum of H_ℓ , stratify each U_ν by s_ν . Lemmas 8.1 and 8.2 bound the incidence dimension by

$$\dim X_L + h_r - \sum_\nu D_{r, s_\nu}.$$

Its image cannot have larger dimension. For fixed r , each term is at least the smallest feasible $D_{r, s}$, so Lemma 8.3 gives

$$\text{codim } \Delta^{(L)} \geq \min\{Lq - p + 1, Lc - 3\} \geq 1.$$

This proves, in particular, generic surjectivity without assuming it.

Proposition 8.4 gives the upper bound $Lc - 3$ and shows that the failure set is nonempty. On a Grassmannian chart the map μ is a $p \times Lq$ matrix over a regular ring. The classical maximal-minor height bound gives codimension at most $Lq - p + 1$ for its nonempty failure locus; see [9, Introduction], with generic rank p and minor size p . The two upper bounds match the lower bound. When $Lc - 3$ is the minimum, the irreducible closed subvariety Σ_L has the maximal dimension of $\Delta^{(L)}$, so it is an irreducible component of its underlying reduced set. If $Lq < p$, the dimension of the domain of μ makes surjectivity impossible. This proves the final equivalence. $\square$

Corollary 8.5 (The base-point locus and the contact count). *For $L = 1$, the locus $\mathcal{B}$ of spaces with a common projective zero is irreducible of codimension $c - 1$, has multiplication corank at least two, and is properly contained in Σ_1 . It is not an irreducible component of $\Delta^{(1)}$. For arbitrary L , the least number of independently chosen c -planes with full product span is*

$$L_{\min} = \left\lceil \frac{2k-1}{\binom{c+1}{2}} \right\rceil.$$

Proof. For a fixed linear factor l , common-base-point spaces form $\text{Gr}(c, lV_{n-1})$. Varying $[l] \in \mathbb{P}^1$ gives an irreducible incidence of dimension $1 + c(k-1-c)$, generically one-to-one because a general residual space has no further common factor. The codimension is $c-1$. Products are divisible by l^2 , whose multiples have dimension $p-2$. The annihilating hyperplane is an evaluation point of $C_n \subset S_n$, proving inclusion in Σ_1 . The codimensions $c-1$ and $c-3$ make that inclusion proper. An irreducible proper subset of the irreducible closed set $\Sigma_1 \subset \Delta^{(1)}$ cannot itself be a component of $\Delta^{(1)}$. The contact count follows from Theorem 5.1 and the dimension obstruction. $\square$

The theorem determines the exact dimension of the entire bad set and one explicit maximal-dimensional component in the stated regime. It does not assert that every component is a secant component, nor that the determinantal scheme is reduced, equidimensional, or everywhere transverse. In particular, the equality case of the two codimension bounds does not exclude additional components of the same dimension.

9. RESIDUAL CONORMALS AND EXACT LOCAL EQUATIONS

We work over $\mathbb{C}$. Put $E_\mu = \bigoplus_\nu \text{Sym}^2 U_\nu$, $M = Lc(k-c)$, $e = L\binom{c+1}{2}$, $p = 2k-1$, and

$$a = e - p + 1, \quad b = Lc - 3.$$

When $e \geq p$, write $\mathcal{D}_L$ for the scheme of maximal minors of $\mu : E_\mu \rightarrow V_{2n}$, and $\mathfrak{J}$ for the incidence $\ell\mu = 0$ in $\mathbb{P}(V_{2n}^*) \times X_L$. The following result is not restricted to a general point or to a stable range of dimensions.

Theorem 9.1 (Residual tangent sequence). *Let $([\ell], U) \in \mathfrak{J}$, let $E = \text{rad } H_\ell$, and put*

$$W_\nu = U_\nu \cap E, \quad d_\nu = \dim W_\nu, \quad R_W = \sum_\nu W_\nu^2 \subset V_{2n}, \quad q = \binom{c+1}{2}.$$

For each ν define

$$B_\nu : \text{Hom}(U_\nu, V_n/U_\nu) \rightarrow \text{Sym}^2 U_\nu^*, \quad B_\nu(A)(u, v) = H_\ell(Au, v) + H_\ell(u, Av).$$

Then $\text{im } B_\nu$ consists exactly of the symmetric forms whose restriction to $W_\nu \times W_\nu$ is zero. There is an exact sequence

$$0 \rightarrow \bigoplus_\nu \ker B_\nu \rightarrow T_{([\ell], U)} \mathfrak{J} \rightarrow R_W^\perp / \mathbb{C}\ell \rightarrow 0. \quad (9.1)$$

In particular

$$\dim T\mathfrak{J} = M + p - 1 - \sum_\nu \left(q - \binom{d_\nu + 1}{2} \right) - \dim R_W. \quad (9.2)$$

If μ_U has corank one, these are also the tangent space and tangent dimension of $\mathcal{D}_L$ at U ; its conormal rank in X_L is

$$\sum_\nu \left(q - \binom{d_\nu + 1}{2} \right) - p + 1 + \dim R_W. \quad (9.3)$$

Proof. The map $V_n/U_\nu \rightarrow U_\nu^*$ induced by H_ℓ has image $(U_\nu/W_\nu)^*$. Indeed its transpose has kernel W_ν ; quotienting by U_ν is permitted because U_ν is isotropic. Choose a complement Z_ν of W_ν in U_ν and lifts of a dual basis of Z_ν^* . The coordinates of $A|_{W_\nu}$ in these lifts prescribe every $W_\nu \times Z_\nu$ entry. The coordinates of $A|_{Z_\nu}$ prescribe every symmetric $Z_\nu \times Z_\nu$ entry, by taking one half of the desired symmetric matrix. This proves the assertion about B_ν , including its rank.

Differentiating $\ell(uv) = 0$ gives

$$\dot{\ell}(uv) + B_\nu(\dot{U}_\nu)(u, v) = 0.$$

It is solvable for the $\dot{U}_\nu$ precisely when $\dot{\ell}$ annihilates all W_ν^2 . Solutions for different contacts are independent. Taking $\dot{\ell}$ modulo $\mathbb{C}\ell$ proves the exact sequence and its dimension. If μ_U has corank one, invert a $(p-1)$ -minor. Row and column operations leave an identity block and one row of a entries. The normalized left kernel vector is uniquely solved by the identity block; the remaining incidence equations are exactly these a entries. Hence the incidence projects isomorphically, as a scheme, to $\mathcal{D}_L$ on this chart. Subtracting the tangent dimension from M gives the conormal rank. $\square$

Proposition 9.2 (Local presentation at every corank). *At a point where μ has corank $\rho \geq 1$, invert a $(p-\rho)$ -minor and write its matrix in blocks as $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$. On this open chart, and also after completion of its local ring,*

$$I_p(\mu) = I_\rho(D - CA^{-1}B). \quad (9.4)$$

The matrix on the right has size $\rho \times (e - p + \rho)$ and vanishes at the given point.

Proof. Multiply by the invertible block matrices eliminating C and B , and then by A^{-1} on the first block. The resulting matrix is $\text{diag}(I_{p-\rho}, D - CA^{-1}B)$, in rectangular block notation. Its maximal minors generate the indicated ideal. Invertible row and column operations preserve determinantal ideals, and localization and completion preserve this identity. $\square$

This is a scheme presentation, not a replacement of its tangent cone by minors of the linear part. Initial ideals can contain additional initial forms obtained from cancellations. Nor does the presentation by itself classify the associated primes of every excess-dimensional germ.

9.1. The rank-two residual morphism. Let $[\ell]$ have rank two and two distinct support points x, y . Choose the two isotropic hyperplanes H_+, H_- of H_ℓ and their intersection $E = gV_{n-2}$, where g vanishes at x, y . Fix a sign pattern ε and consider the open incidence on which $U_\nu \subset H_{\varepsilon_\nu}$ and $U_\nu \not\subset E$.

Proposition 9.3 (Dominance with affine fibres). *For fixed $[\ell]$ and choices of its hyperplanes, the morphism*

$$(U_\nu)_\nu \mapsto (g^{-1}(U_\nu \cap E))_\nu \quad \text{to} \quad \text{Gr}(c-1, V_{n-2})^L \quad (9.5)$$

is smooth and surjective. It is locally an affine-space bundle of relative dimension $L(k-c-1)$. The same assertion holds relatively as the distinct support points and their two hyperplanes vary, after trivializing the indicated bundles.

Proof. For a prescribed $W_\nu = gK_\nu$, an admissible U_ν is the inverse image of a line in $H_{\varepsilon_\nu}/W_\nu$ projecting isomorphically to H_{ε_ν}/E . Its choices form a torsor under

$$\text{Hom}(H_{\varepsilon_\nu}/E, E/W_\nu),$$

an affine space of dimension $k-c-1$. Local choices of a splitting trivialize this torsor. Intersections with E have constant dimension $c-1$, so the assignment is a morphism. Products give the result. $\square$

Corollary 9.4 (Exact singularity recursion on a signed stratum). *At a rank-two point as above, suppose that μ_U has corank one. Let μ'_K be multiplication from $\bigoplus_\nu \text{Sym}^2 K_\nu$ to V_{2n-4} and put $\delta = \dim \text{coker } \mu'_K$. Then*

$$\dim T_U \mathcal{D}_L = M - b + \delta. \quad (9.6)$$

If $4 \leq c < k$, $e \geq p$ and $b \leq a$, this point is smooth in $\mathcal{D}_L$ if and only if μ'_K is onto. In this range the singular set on the stated corank-one open stratum is exactly the inverse image of the residual multiplication failure locus.

Proof. Here $d_\nu = c - 1$ and $R_W = g^2 \text{im } \mu'_K$, of dimension $p - 4 - \delta$. Equation (9.2) gives the displayed formula. The signed incidence has image of dimension $M - b$: a general first c -plane in its secant hyperplane determines that hyperplane uniquely, since the locus also lying in a second secant hyperplane has dimension at most $3 + c(k - 2 - c) < c(k - 1 - c)$. On the corank-one open set the local signed incidence is embedded by the incidence isomorphism. Thus its local dimension is $M - b$. Theorem 5.1 bounds the dimension of the entire failure locus by $M - b$ when $b \leq a$. Consequently its local dimension here is $M - b$, and equality of tangent and local dimensions is equivalent to $\delta = 0$. $\square$

9.2. Projection and commutative algebra.

Proposition 9.5 (From rank incidences to components). *Stratify the annihilator incidence by the rank r of H_ℓ and by $s_\nu = \dim(U_\nu / (U_\nu \cap \text{rad } H_\ell))$. Every irreducible component C of the support of $\mathcal{D}_L$ is dominated by the closure of an irreducible piece of one of these finitely many strata. In particular a stratum of dimension less than $\dim C$ cannot dominate C . If a dominating incidence has dimension $\dim C$, and its generic fibre contains a nonempty open subset of the projective annihilator space, then that annihilator space is a point and its projection is birational in characteristic zero.*

Proof. The projective incidence maps onto the failure locus. Intersect the images of its finitely many locally closed strata and their irreducible pieces with C . These constructible sets cover C ; at least one is dense. The dimension theorem gives the first assertion and implies that the generic fibre of an equal-dimensional dominating piece is zero-dimensional. The projective annihilator space is the projectivization of a vector space, not an arbitrary finite collection. A nonempty open subset of a positive-dimensional projective space has positive dimension. Thus the open-fibre hypothesis forces the projective space to be a point. On a corank-one chart the incidence is the graph of its uniquely solved annihilator. This proves birationality, including the scheme-theoretic generic-point assertion. $\square$

Proposition 9.6 (Expected grade and absence of embedded primes). *Suppose $4 \leq c < k$, $e \geq p$, and $a \leq b$. On every nonempty affine Grassmannian chart meeting $\mathcal{D}_L$, its maximal-minor ideal has height and grade a . The Eagon–Northcott complex resolves its quotient, which is Cohen–Macaulay of pure codimension a and has no embedded associated primes.*

Proof. The chart ring is regular. The global codimension theorem gives height at least a , whereas the maximal-minor height bound gives height at most $e - p + 1 = a$. In a regular ring grade equals height. The Eagon–Northcott grade criterion gives a free resolution of length a and the Cohen–Macaulay conclusion [8, 19]. Localization gives the same conclusion at every prime over the ideal. A Cohen–Macaulay quotient of this expected dimension is unmixed and has no embedded associated primes. No independence assumption on the matrix entries is used. $\square$

Proposition 9.7 (The reducedness step). *In the situation of Proposition 9.6, if $\mathcal{D}_L$ is generically reduced on each of its irreducible components, then it is reduced. If its support is also irreducible, it is integral.*

Proof. A Noetherian ring without embedded associated primes satisfies S_1 . Generic reducedness is R_0 . The criterion $R_0 + S_1$ for reducedness therefore applies [24, Tag 031R]. Alternatively a nonzero nilradical would have an associated prime; without embedded primes this would be a minimal prime, contradicting generic reducedness. The last assertion is the definition of an integral scheme. $\square$

10. ORIENTATION COVERS AND DESCENT OF ISOTROPIC FAMILIES

We give the descent statements used in the component theorem explicitly. They distinguish connectedness of a cover from geometric irreducibility of a fibre, and also distinguish one maximal-isotropic factor from a product of such factors.

Proposition 10.1 (Relative orthogonal descent). *Let S be a smooth integral complex variety and let $(\mathcal{V}, q)$ be a nondegenerate quadratic bundle of rank $2s$, with values in a line bundle. Its orientation cover $\tilde{S} \rightarrow S$ is the degree-two étale cover whose local equation, after choosing frames, is $z^2 = (-1)^s \det q$. Changes of frames glue these local covers. Suppose this cover is connected. The relative maximal orthogonal Grassmannian $\text{OGr}_S(s, \mathcal{V})$ is integral. More generally, the product of L such Grassmannians for the same quadratic bundle has 2^{L-1} irreducible components, indexed by $\{+, -\}^L$ modulo simultaneous sign change. Each component is smooth over S . Adding relative nonmaximal orthogonal Grassmannian factors does not change this count.*

Proof. If a frame of $\mathcal{V}$ changes by A , its determinant changes by $(\det A)^2$. If the frame of the value line changes by u , the determinant changes by u^{-2s} . Thus the square-root equations are compatible with the transition functions of $\det \mathcal{V}^* \otimes (\text{value line})^s$. Their zero-free right-hand side makes the cover étale. After an étale base change the form has a hyperbolic basis $e_1, \dots, e_s, f_1, \dots, f_s$.

Over an algebraically closed field a maximal isotropic plane belongs to one of two families, distinguished by the parity of its intersection dimension with $\text{span}(e_1, \dots, e_s)$. A Witt basis sends any such plane to the reference plane by an orthogonal transformation. The transformations of determinant one preserve this parity and act transitively on each family; interchanging e_1 and f_1 reverses it. The special orthogonal group is connected: in a hyperbolic basis its elementary orthogonal root subgroups are affine lines, its diagonal torus is a product of multiplicative groups, and these generate the group by the usual elimination of a Witt basis. Therefore each of its two orbits is irreducible. Equivalently, the graph charts within a family are alternating matrices and are translated under this connected group. The same Witt-basis elimination gives a single irreducible orbit for isotropic planes of dimension strictly less than half the rank.

The two families are consequently locally constant étale labels, with the determinant character describing their interchange. The orientation cover kills this character. Over $\tilde{S}$ each family is a smooth proper relative variety with geometrically irreducible fibres. It is integral: smoothness gives reducedness, every component dominates the integral base, and the geometric generic fibre has only one component. Here domination follows also from local smoothness and properness: each component is open and closed in the smooth total space, so its nonempty image is open and closed in the connected base. The nontrivial deck transformation exchanges the two families. Their images in the descended space are therefore the same irreducible image, which is the whole space. This proves integrality for one factor.

For L factors the geometric components over $\tilde{S}$ are the 2^L products of these integral family bundles. The deck action sends ε to $-\varepsilon$. The orbit unions descend to the irreducible components: the image of any one integral product is irreducible and equals the image of its entire orbit, and distinct orbits have distinct geometric generic fibres. There are 2^{L-1} orbits. Smoothness descends étale locally. Nonmaximal factors have geometrically irreducible fibres and add no labels. $\square$

Proposition 10.2 (The five exceptional orientation bases). *For each exceptional tuple of Theorem 11.1, the incidence containing the full radical and a maximal isotropic quotient is irreducible. For the full-rank case $(c, k, L) = (7, 14, 1)$ the nondegenerate isotropic incidence is irreducible as well.*

Proof. For an exceptional rank $r = 2s < k$, work first over the dense ordered support chart

$$T_r = \{(x_1, \dots, x_r, \alpha_1, \dots, \alpha_{r-1}) : x_i \neq x_j, \alpha_i \neq 0\}, \quad \alpha_r = 1.$$

This is a smooth integral open subset of $\mathbb{A}^r \times \mathbb{G}_m^{r-1}$. It parameterizes the projective functionals $[\ell] = [\sum_i \alpha_i \text{ev}_{x_i}]$. Evaluation identifies the quotient by the radical with $\mathbb{C}^r$, and the induced quadratic form is $\text{diag}(\alpha_1, \dots, \alpha_r)$. In the function field $\mathbb{C}(x_1, \dots, x_r, \alpha_1, \dots, \alpha_{r-1})$, the product $\prod_{i < r} \alpha_i$ is not a square: its valuation at $\alpha_1 = 0$ is one. This valuation is a function-field valuation; the boundary $\alpha_1 = 0$ need not belong to T_r . The orientation cover obtained by adjoining its square root is therefore integral and connected. Proposition 10.1 applies.

The radical is gV_{k-r-1} with $g = \prod_i (t - x_i)$. Containing this radical identifies the exceptional plane incidence with the maximal orthogonal Grassmannian of the quotient. There is one such factor in each of the five cases. The ordered chart maps dominantly to the corresponding exact-rank base. Its generic fibres merely relabel the support: the radical determines g , its distinct roots recover the unordered points, and a Vandermonde system recovers the weights up to common scale. Taking the unordered quotient cannot split the irreducible image of this incidence. Its closure is the exceptional incidence claimed in the theorem. Colliding-support points do not create another component of this closure.

In the remaining full-rank case the universal quadratic form has even rank fourteen. The determinant of the universal Hankel matrix has the irreducible corank-one Hankel hypersurface as its zero divisor, with multiplicity one. The irreducibility follows from the Hankel maximal-minor description in Section 8; simplicity can also be checked at a corank-one form represented by thirteen distinct evaluations. Its kernel is generated by a nonzero polynomial v ; the derivative of the determinant in a Hankel direction ℓ is a nonzero scalar multiple of $\ell(v^2)$, which is not identically zero. On an affine projective chart meeting its generic point this simple prime divisor proves nonsquareness in the function field. Scaling the representative of a rank-fourteen form changes the determinant by a fourteenth power, hence by a square, so this is the projective orientation cover. It is connected and Proposition 10.1 gives the assertion. $\square$

11. COMPONENTS IN ALL DIMENSIONS

We retain the entire range $4 \leq c < k$ and $Lq \geq p$ of Theorem 5.1. Let Φ_L be the closure of the image of the incidence with nonsingular H_ℓ . Let Σ_ε denote the signed secant image: for distinct x, y and $t \neq 0$ its points satisfy

$$U_\nu \subset \ker(\text{ev}_x + \varepsilon_\nu t \text{ev}_y), \quad \varepsilon \in \{+1, -1\}^L / \{\pm 1\}.$$

Only in the five triples in Table 1 do we need an additional notation. For those triples, $\mathcal{T}_{c,k}$ is the closure of the image of the rank- r incidence indicated in the table. Equivalently, on its dense distinct-support open set,

$$E = \text{rad } H_\ell \subset U, \quad U/E \text{ is a maximal isotropic subspace of } V_n/E.$$

Thus these exceptions have one uniform geometric description; they are not five unrelated numerical accidents.

Theorem 11.1 (All-dimension component theorem). *Suppose $4 \leq c < k$ and $Lq \geq p$.*

- (1) *If $a < b$, the scheme $\mathcal{D}_L$ is integral and Cohen–Macaulay of codimension a . Its support is Φ_L , except at $(c, k, L) = (5, 8, 1), (6, 11, 1)$, where its support is $\mathcal{T}_{5,8}, \mathcal{T}_{6,11}$, respectively.*

c	k	L	(a, b)	r	$\dim E$	$\dim(U/E)$
5	7	1	(3, 2)	4	3	2
5	8	1	(1, 2)	6	2	3
6	10	1	(3, 3)	8	2	4
6	11	1	(1, 3)	10	1	5
7	13	1	(4, 4)	12	1	6

TABLE 1. All nonendpoint equality incidences.

- (2) If $a = b$, the scheme is reduced and Cohen–Macaulay of pure codimension a . Its components are Φ_L and the 2^{L-1} varieties Σ_ε , except at $(6, 10, 1)$, $(7, 13, 1)$, where Φ_L is replaced by $\mathcal{T}_{6,10}, \mathcal{T}_{7,13}$.
- (3) If $b < a$, the maximal-dimensional components are exactly the 2^{L-1} signed secant varieties, except at $(5, 7, 1)$, where $\mathcal{T}_{5,7}$ is one additional maximal-dimensional component. Their codimension is b .

Every component listed above has general multiplication corank one and is generically smooth in the maximal-minor scheme. Each of its indicated annihilator incidences is birational onto it. The generic Hankel rank is k on Φ_L , two on Σ_ε , and the rank in Table 1 on $\mathcal{T}_{c,k}$. The signed components are rational. In the range $a \leq b$,

$$[\mathcal{D}_L] = c_a \left(\left(\bigoplus_{\nu} \text{Sym}^2 \mathcal{S}_{\nu} \right)^* \right) \quad \text{in } A^a(X_L). \quad (11.1)$$

The third assertion concerns maximal-dimensional components, not all associated primes in the excess range. The theorem removes the stable restriction from the component and expected-codimension scheme assertions; it does not identify distinct notions of completeness.

11.1. The complete equality analysis.

Lemma 11.2 (All endpoint ties). *Among the rank incidences in the proof of Theorem 5.1, an incidence of dimension at least $M - \min(a, b)$ is either the rank-two incidence with every $s_{\nu} = 1$, the full-rank incidence with every $s_{\nu} = c$, or one of the five incidences in Table 1. The dimension in each listed case is exactly $M - \min(a, b)$ when its corresponding endpoint is minimal.*

Proof. Use $D_{r,s}$ and h_r in Lemmas 8.1 and 8.2. For $r < k$ and $s = 0$ the cost is $(Lc - 2)r + 1 > b$. For $1 \leq s \leq c - 2$, minimization in $r \geq 2s$ uses the nonnegative coefficient $L(c - s) - 2$ and gives

$$g(s) = 1 + s(Lc - 4) - Ls(s - 1)/2, \quad g(1) = b, \quad g(c - 2) - b = (c - 3)\{L(c + 2)/2 - 4\}.$$

If $L \geq 2$, the last difference is strictly positive, and strict concavity excludes the intervening integers. The coefficient for $s = 1$ is positive, so equality there requires $r = 2$. For $s = c - 1$, $L \geq 2$, the cost at its minimum exceeds b by $(c - 2)\{L(c + 1)/2 - 4\} > 0$. For $s = c$ and $r < k$, the cost is at least $a + 1$. Thus there are no extra ties for $L \geq 2$.

It remains to take $L = 1$. Saturation gives $k \leq (q + 1)/2$. For $c = 4$, this forces $k = 5$ and feasibility permits only $s \leq 1$ when $r < k$. For $c = 5$, the values at $s = 1, 2, 3$ are 2, 2, 1. At $s = 2$ equality requires $r = 4$, and feasibility together with the minimal endpoint gives $k = 7$. At $s = 3$ the coefficient of r is zero; feasibility forces $r = 6, k = 8$. For $c = 6$, the only further lower-endpoint tie is $s = 4, r = 8, k = 10$. For $c \geq 7$, strict concavity and the displayed endpoint difference exclude every $2 \leq s \leq c - 2$.

For $s = c - 1$ the cost is decreasing in r and is bounded below by $a + k - 2c + 1$. Feasibility requires $k \geq 2c - 1$. When $k \geq 2c$, this is strictly greater than a . When $k = 2c - 1$, feasibility

forces $r = 2c - 2$, and

$$a - b = (c - 7)(c - 2)/2.$$

Saturation permits this case for $c \geq 6$; the minimal-endpoint ties are precisely $c = 6$ and $c = 7$, giving (6, 11) and (7, 13). For $c \geq 8$ the cost a is larger than b . The case $s = c$ was already excluded. This proves the one-contact list.

Finally, if the contacts have different s_ν , write their total cost as the average of $LD_{r,s_\nu} - h_r$. Every summand is at least $\min(a, b)$ by Lemma 8.3; equality of the average requires equality of every summand. The preceding analysis shows that for $L \geq 2$ only the identical endpoint choices remain. The exceptional cases all have $L = 1$. $\square$

Lemma 11.3 (Three-dimensional residual subseries). *For $K \geq 3$ and $L \geq 1$, general three-planes in a K -dimensional binary space have full quadratic product span whenever $6L \geq 2K - 1$, provided either $K = 3$ or $L \geq 2$.*

Proof. For $K = 3$ the plane is the entire binary space. Otherwise stratify the annihilator incidence as before, now with $c = 3$ and $L \geq 2$. For $r < K$, the costs for $s = 0, 1, 2, 3$ are respectively

$$(3L - 2)r + 1, \quad (2L - 2)r - L + 1, \quad (L - 2)r + L + 1, \quad 6L - 2r + 1.$$

On their feasible ranges they are positive: use $r \geq 2$ for $s = 1$, $r \geq 4$ for $s = 2$, and $r \leq K - 1$ for $s = 3$. The full-rank cost, when feasible, is $6L - 2K + 2 \geq 1$. The projective incidence therefore cannot dominate the Grassmannian product. This proves generic surjectivity. $\square$

11.2. Irreducibility and the exceptional geometry.

Lemma 11.4 (Signed incidences in the full range). *For $c \geq 4$, the 2^{L-1} signed secant images are distinct, irreducible and rational of codimension b . A general point of one admits exactly one rank-two projective annihilator.*

Proof. A general secant hyperplane has a unique unordered decomposition into two evaluations: evaluations at four distinct points are independent for $n \geq 3$, and the limiting assertion follows by Hermite interpolation. Its partner hyperplane and its rank-two quadratic annihilator are therefore determined. The secant parameter space is rational, being birational to a projective-line bundle over $\text{Sym}^2 \mathbb{P}^1$. Normalize the first sign to be positive. The incidence is a product of relative Grassmannians of dimension $3 + Lc(k - 1 - c) = M - b$, rational over this rational base.

For a fixed general first hyperplane h , planes admitting a second secant hyperplane have dimension at most $3 + c(k - 2 - c)$ in $\text{Gr}(c, \ker h)$, strictly less than $c(k - 1 - c)$. Thus the first plane recovers h generically. A general other plane lies in only its chosen partner, proving distinctness of the sign sectors and birationality. Every rank-two isotropic plane is contained in one of its isotropic hyperplanes. The exceptional confluent secants have smaller base dimension. These facts also give uniqueness of the general rank-two annihilator. $\square$

Lemma 11.5 (Orthogonal incidences and their orientation covers). *In the expected-codimension range, the nondegenerate incidence, when feasible, is irreducible of dimension $M - a$ and contains points outside all the signed images. Each exceptional incidence in Table 1 is irreducible and its image also contains points outside the signed images.*

Proof. If $2c < k$, the nondegenerate incidence is smooth over the irreducible open set of nonsingular Hankel forms, with geometrically irreducible fibres $\text{OGr}(c, k)^L$. The dimension of each factor is $c(k - c) - q$. For a fixed nonsingular form, isotropic c -planes in a hyperplane have codimension at least c in this Grassmannian: if the restricted form has a one-dimensional radical, separate planes containing that radical from those avoiding it. Their codimension drops are respectively $k - c - 1 \geq c$ and c . Varying a secant hyperplane contributes three parameters, fewer than c . Hence the first plane can be chosen outside every secant hyperplane.

The inequalities $a \leq b$ and $e \geq p$ imply $2c < k$, except for the four expected-codimension entries of Table 1 and $(c, k, L) = (7, 14, 1)$. This follows at once for $c \geq 8$ from $2k \geq L \binom{c}{2} + 5$; for $4 \leq c \leq 7$ substitution gives exactly these five possibilities. In the four table entries the full-rank incidence is empty. For $(7, 14, 1)$ its maximal orthogonal fibres have two components, but the total incidence is irreducible. Indeed the orientation double cover adjoins a square root of the determinant of the universal Hankel form. This determinant is not a square in the function field: its zero divisor is the irreducible rank-at-most-13 Hankel hypersurface, and it vanishes simply at a general corank-one point. Simplicity also follows by perturbing that point by one further evaluation nonzero on its kernel. The connected orientation cover exchanges the two maximal orthogonal families. Over it each family has geometrically irreducible fibres, which proves irreducibility of the total incidence. In a maximal orthogonal Grassmannian the condition of containing a fixed isotropic line has codimension $c - 1 = 6$; the same three-parameter secant argument gives a nonsecant first plane.

For an exceptional incidence write $r = 2(k - c)$ and $d = 2c - k$. On the dense distinct-support open set,

$$\ell = \sum_{i=1}^r \alpha_i \operatorname{ev}_{x_i}^{(2n)}, \quad E = gV_{d-1}, \quad V_n/E \simeq \mathbb{C}^r,$$

with quotient form $\operatorname{diag}(\alpha_i)$ and all $\alpha_i \neq 0$. The product $\prod_i \alpha_i$ is nonsquare in the function field of the ordered support and projective weight parameter space (normalize the last weight to one). Thus its orientation cover is connected. The same argument for maximal orthogonal fibres proves irreducibility of this incidence. Its closure defines the incidence used here; lower-dimensional support collisions do not add another generic family.

To find a point outside the secant images, first suppose $d \geq 2$. A secant functional annihilating gV_{d-1} must have both its support points among the roots of g , since evaluations of V_{d-1} at two distinct points are independent. A confluent secant contributes only a pure evaluation, because g has simple roots. In the nondegenerate quotient an admissible hyperplane must have isotropic dual vector. Each of the finitely many secant lines through support points contains only two such vectors; a pure evaluation has anisotropic dual. A maximal isotropic s -plane containing any fixed isotropic line is a proper subvariety of codimension $s - 1 > 0$. Avoid this finite union. If $d = 1$, the secant functionals annihilating g form a set of dimension at most two. Here $s = 5$ or 6 , and $s - 1 > 2$, so the same incidence bound again leaves a nonempty complement. This proves the assertion. $\square$

Lemma 11.6 (A corank-one point in the excess exception). *The image $\mathcal{T}_{5,7}$ has dimension $M - b$ and general multiplication corank one.*

Proof. Dehomogenize and choose a monic square-free polynomial g of degree four. In V_6 put

$$U = \operatorname{span}\{1, t, g, tg, t^2g\}.$$

For its four roots x_i , the nonzero barycentric weights $\alpha_i = 1/g'(x_i)$ satisfy $\sum_i \alpha_i x_i^j = 0$ for $j = 0, 1, 2$. Thus $\ell = \sum_i \alpha_i \operatorname{ev}_{x_i}^{(12)}$ has Hankel rank four, radical gV_2 , and annihilates U^2 . The twelve products

$$1, t, t^2, g, tg, t^2g, t^3g, g^2, tg^2, t^2g^2, t^3g^2, t^4g^2$$

have distinct degrees, hence are independent. They span a hyperplane of V_{12} . The annihilator is unique, so the incidence projection is an isomorphism on a neighbourhood in its corank-one open set. Its image has the incidence dimension $M - b$. This example is not in a secant hyperplane. Since it contains gV_2 , any secant annihilator would have both support points among the roots of g , by the argument in Lemma 11.5. Evaluations of $1, t$ at two distinct such roots are independent; a pure evaluation is also excluded. $\square$

11.3. Proof of the component theorem.

Proof of Theorem 11.1. We use Propositions 10.1 and 10.2 for the descent of maximal orthogonal families; in particular a connected orientation cover is not being confused with geometric irreducibility of its fibres.

Assume first $a \leq b$. Proposition 9.6 makes $\mathcal{D}_L$ pure of dimension $M - a$. By Proposition 9.5 and Lemma 11.2, its components must be dominated by the listed equal-dimensional incidences. In the strict case only the nondegenerate incidence, or the specified exceptional replacement, can dominate. It is irreducible, so the nonempty failure scheme has just that support. At the wall the signed images already have the component dimension. The nonsecant point of Lemma 11.5, purity, and the same rank-gap argument force one further component, dominated by the nondegenerate incidence or its exceptional replacement. There can be no others.

For each nondegenerate or exceptional component, its incidence and image have the same dimension. In the nondegenerate case full rank is open in its projective annihilator space. In an exceptional case the rank $r = 2(k - c)$ is the largest rank permitting an isotropic c -plane: $\dim U \leq k - r + \lfloor r/2 \rfloor$ excludes every larger rank. It too is an open rank condition in the annihilator space. A zero-dimensional nonempty open subset of a projective space forces that space to be a point. This proves corank one and birationality.

When $b < a$, the same incidence-to-component argument lists all possible maximal-dimensional components. The signed images have the required dimension, and Lemma 11.6 supplies the additional component at $(5, 7, 1)$. No other rank incidence has sufficient dimension.

We verify generic corank one on the signed components without assuming it in this dimension argument. At a general signed point the rank-gap lemma excludes all strictly smaller incidences. If its annihilator space had positive dimension and generic rank greater than two, its open maximum-rank part would have positive dimension. The only remaining possible incidences have dimension $M - b$, so cannot have a positive-dimensional fibre over a dense subset of a signed component of that same dimension. If the generic rank were two, the entire open part would contradict uniqueness of the rank-two annihilator in Lemma 11.4. These alternatives exhaust a projective space containing its known rank-two point.

The signed residual planes are general by Proposition 9.3. Since $b \leq a$ is exactly

$$L \binom{c}{2} \geq 2k - 5,$$

their multiplication is onto. If $k = c + 1$, each residual plane is the entire residual binary space, whose square spans the target. Otherwise apply the earlier codimension theorem at $(k - 2, c - 1)$ when $c \geq 5$. For $c = 4$ use Lemma 11.3; if $L = 1$, saturation here forces $k = 5$ and the residual space is the entire three-dimensional binary space. Corollary 9.4 therefore proves generic scheme smoothness on the signed components.

For a nondegenerate component all W_ν are zero, so Theorem 9.1 gives tangent dimension $M - a$. For an exceptional component $L = 1$, $W = E = gV_{d-1}$, and $\dim W^2 = 2d - 1$. Its tangent dimension is

$$M - a + \binom{d+1}{2} - (2d - 1) = M - a + (d - 1)(d - 2)/2.$$

For $d = 1, 2$ this is $M - a$. For the excess exception $d = 3$, it is $M - a + 1 = M - b$. In each case it is exactly the local dimension, proving generic scheme smoothness. Proposition 9.7 now proves reducedness, and integrality in the strict case, of the entire expected-codimension scheme.

Finally apply the Thom–Porteous formula to the dual map with source $V_{2n}^* \otimes \mathcal{O}$ of rank p and target $(\bigoplus_\nu \text{Sym}^2 \mathcal{S}_\nu)^*$ of rank e , at rank bound $p - 1$ [16, 11]. Expected codimension has

been proved and gives (11.1). Reducedness makes this the sum of the component cycles with multiplicity one. No excess-codimension cycle formula is asserted. $\square$

12. SIGNED INTERSECTIONS AND THE RESIDUAL WALL

This section distinguishes three objects: the incidence over the rank-two base, its reduced normalization, and its image in the full multiplication failure scheme. They agree neither globally nor in all scheme structures.

12.1. Normalization and the sign incidence complex. Let B_2° be the open set of rank-two projective Hankel forms with two distinct support points and nonzero weights. Choosing one of the two isotropic hyperplanes gives a finite etale double cover $\tilde{B}_2^\circ \rightarrow B_2^\circ$. On that cover the hyperplane bundles H_+, H_- and their common radical bundle E are labelled. Denote by $\mathfrak{I}_2^\circ$ the restriction of $\mathfrak{I}$ to this base.

Theorem 12.1 (Normalization and all same-annihilator signed intersections over the split rank-two base). *After this double-cover base change, the normalization of $(\mathfrak{I}_2^\circ)_{\text{red}}$ is the disjoint union of the smooth relative varieties*

$$Y_\varepsilon = \prod_{\nu=1}^L \text{Gr}(c, H_{\varepsilon_\nu}), \quad \varepsilon \in \{+, -\}^L.$$

Descent identifies ε with $-\varepsilon$, giving 2^{L-1} irreducible components before the cover.

For any nonempty collection $\mathcal{A}$ of sign patterns let J be the set of contacts at which those patterns do not all agree. Their scheme-theoretic intersection as reduced components over the labelled base is

$$\bigcap_{\varepsilon \in \mathcal{A}} Y_\varepsilon = \prod_{\nu \in J} \text{Gr}(c, E) \times \prod_{\nu \notin J} \text{Gr}(c, H_{\varepsilon_\nu}). \quad (12.1)$$

When nonempty it is smooth of dimension $M - b - c|J|$ and has codimension $c|J|$ in each Y_ε . These are also the local projected intersections wherever the multiplication map has corank one and its annihilator belongs to B_2° .

Proof. For a split rank-two form its quadratic restriction is a product of two linear functionals. A vector subspace is isotropic exactly when one of these functionals vanishes identically on it: the polynomial ring on that subspace is a domain. The connected cover and its deck action are also established algebraically in Proposition 13.1 with $m = 2$. This identifies the reduced incidence as the union of the displayed Y_ε . Each is smooth and normal. Their disjoint union maps finitely to the union, is birational on every component, and hence is its normalization. The deck transformation simultaneously interchanges the two hyperplanes; its action on sign patterns has no fixed pattern. The cover is connected, as is also seen from the parameterization by a secant hyperplane and its uniquely recovered partner. Thus its orbits give the stated component count.

The condition $U \subset H_+$ and $U \subset H_-$ is exactly $U \subset E$. On any frame chart these are independent linear containment equations; adding the ideals of the reduced components therefore gives precisely the Grassmannians in (12.1), with their reduced scheme structures. Replacing $\text{Gr}(c, k-1)$ by $\text{Gr}(c, k-2)$ lowers dimension by c at each differing contact. This proves the formula, including emptiness if a required c -plane cannot fit in E . The corank-one incidence isomorphism transfers the local assertion to the projected scheme. $\square$

The restriction on the last statement is essential. Two signed images can also meet through different annihilators at a higher-corank point, or over a support collision outside B_2° . Formula (12.1) does not identify such points with a single labelled annihilator.

Proposition 12.2 (The embedded structure of a rank-two fibre). *Fix a split rank-two form and one contact with $U \subset E$. Up to smooth local factors, its full isotropy fibre has coordinate ring*

$$R/J_c, \quad R = \mathbb{C}[a_1, \dots, a_c, b_1, \dots, b_c], \quad J_c = (a_i b_i, a_i b_j + a_j b_i \ (i < j)). \quad (12.2)$$

Its radical is $(a_1, \dots, a_c) \cap (b_1, \dots, b_c)$. For $c \geq 2$ its nilradical has dimension $\binom{c}{2}$ as a $\mathbb{C}$ -vector space, is annihilated by all a_i, b_i , and squares to zero. Its associated primes are exactly

$$(a_1, \dots, a_c), \quad (b_1, \dots, b_c), \quad (a_1, \dots, a_c, b_1, \dots, b_c).$$

For several contacts the fibre is the product of the corresponding isotropy fibres, with independent contact coordinates.

Proof. Choose quotient coordinates so that the rank-two quadratic form is $2ab$. In a Grassmannian frame chart at $U \subset E$, its two quotient coordinates on the i th frame vector are a_i, b_i . The remaining coordinates are unconstrained radical directions. Restricting the bilinear form to all pairs of frame vectors gives exactly J_c . Its zero set is the union of the two coordinate c -planes, by the factorization argument above, proving the radical assertion.

Modulo J_c , the mixed quadratic monomials satisfy $a_i b_i = 0$ and $a_i b_j = -a_j b_i$. Their classes therefore form $\wedge^2 \mathbb{C}^c$, with no further degree-two relations. Every mixed cubic vanishes: for example

$$a_i a_k b_j = -a_i a_j b_k = a_j a_k b_i = -a_k a_i b_j,$$

so it is zero in characteristic zero; the argument for one a and two b 's is identical. All mixed monomials of higher degree vanish as well. This proves the assertions about the nilradical N . In the exact sequence

$$0 \longrightarrow N \longrightarrow R/J_c \longrightarrow R/\sqrt{J_c} \longrightarrow 0,$$

N has only the origin as associated prime, whereas the reduced quotient has its two minimal primes and no embedded prime. The associated-prime containment for a short exact sequence permits no others, and all three occur when $c \geq 2$. The equations for different contacts use disjoint variables, proving the product presentation. $\square$

Proposition 12.3 (Primary decomposition and Hilbert series of the fibre). *In the notation of (12.2), let $\mathfrak{a} = (a_1, \dots, a_c)$, $\mathfrak{b} = (b_1, \dots, b_c)$, and $\mathfrak{m} = \mathfrak{a} + \mathfrak{b}$. For $c \geq 2$ an irredundant primary decomposition is*

$$J_c = \mathfrak{a} \cap \mathfrak{b} \cap (J_c + \mathfrak{m}^3). \quad (12.3)$$

The last ideal is $\mathfrak{m}$ -primary, and the Hilbert series is

$$\text{Hilb}_{R/J_c}(t) = \frac{2}{(1-t)^c} - 1 + \binom{c}{2} t^2. \quad (12.4)$$

Proof. The intersection $\mathfrak{a} \cap \mathfrak{b}$ is generated by all mixed monomials $a_i b_j$. Every mixed monomial of degree at least three belongs to J_c , by the cubic calculation in the preceding proof. In degree two, intersection with $J_c + \mathfrak{m}^3$ leaves exactly the degree-two part of J_c . Homogeneity proves (12.3) degree by degree. The quotient by $J_c + \mathfrak{m}^3$ is an Artinian local ring with maximal ideal the image of $\mathfrak{m}$; every zero divisor is nilpotent. Thus this ideal is $\mathfrak{m}$ -primary. Neither minimal prime can be omitted: a_1^3 witnesses the need for $\mathfrak{b}$ and b_1^3 for $\mathfrak{a}$. The embedded component cannot be omitted because $a_1 b_2$ survives modulo J_c when $c \geq 2$. The reduced quotient has in each positive degree the direct sum of the pure a - and pure b -monomials, and in degree zero one common constant. The nilradical adds exactly $\binom{c}{2}$ degree-two classes. This proves (12.4) and verifies algebraically that the three primes listed above are all the associated primes. For $c = 1$ the third component is redundant and $J_1 = (a_1 b_1)$ is already reduced. $\square$

This embedded prime is a statement about a fibre with its annihilator fixed. It does not assert an embedded prime of $\mathcal{D}_L$. In particular a reduced total scheme can have these nonreduced fibres.

12.2. Ordinary double crossings at the residual wall. For the next theorem suppose

$$c \geq 8, \quad k \geq 2c + 1, \quad a = b. \quad (12.5)$$

The parity in $L\left(\frac{c}{2}\right) = 2k - 5$ in fact forces $c \geq 10$ in this range. On the signed open set of Proposition 9.3, the residual map is square:

$$\mu'_K : \bigoplus_{\nu} \text{Sym}^2 K_{\nu} \longrightarrow V_{2n-4}, \quad L\left(\frac{c}{2}\right) = 2k - 5.$$

Let Z_{ε} be the inverse image of its determinant divisor. It is a divisor in the signed incidence, rather than a new arbitrary choice of a discriminant.

12.3. Relative determinant and the normal derivative. In this subsection $B = \tilde{B}_2^{\circ}$ is the connected base labelling the two isotropic hyperplanes, and $\mathcal{R}$ is their common radical bundle. We impose the wall hypotheses (12.5). These relative statements provide the bundle identifications and the nonemptiness arguments for the node normal form.

Proposition 12.4 (The relative residual divisor). *There is a subbundle $\mathcal{T} \subset V_{2n} \otimes \mathcal{O}_B$ with fibre $g^2 V_{2n-4}$ and rank $h = p - 4$. On*

$$P = \prod_{\nu=1}^L \text{Gr}_B(c-1, \mathcal{R})$$

let $\mathcal{W}_{\nu}$ denote the tautological bundles. The determinant of residual multiplication

$$\tau : \bigoplus_{\nu} \text{Sym}^2 \mathcal{W}_{\nu} \longrightarrow \mathcal{T}$$

is a section of the line bundle

$$\det \mathcal{T} \otimes \left(\det \bigoplus_{\nu} \text{Sym}^2 \mathcal{W}_{\nu} \right)^{-1}. \quad (12.6)$$

Its zero scheme Z_P is an integral reduced Cartier divisor in P . For each sign pattern the signed open incidence $S_{\varepsilon} \rightarrow P$ is a smooth affine-space torsor. Its inverse image $Z_{\varepsilon} = S_{\varepsilon} \times_P Z_P$ is likewise an integral reduced divisor. These assertions hold on the entire distinct-support labelled base, not only after fixing the two points.

Proof. The unordered pair of support points gives a line subbundle $\mathcal{G} \subset V_2 \otimes \mathcal{O}_B$. Multiplication gives isomorphisms

$$\mathcal{R} = \mathcal{G} \otimes V_{n-2}, \quad \mathcal{T} = \mathcal{G}^{\otimes 2} \otimes V_{2n-4} \subset V_{2n} \otimes \mathcal{O}_B.$$

These formulas are homogeneous and so include a support point at infinity. The source and target of τ have the same rank h at the wall, proving (12.6). Trivializing $\mathcal{G}$ identifies P with the product of the base chart and $\text{Gr}(c-1, V_{n-2})^L$, and identifies the determinant zero scheme with the same product for the ordinary residual multiplication divisor. By Theorem 11.1 at $(k', c') = (k-2, c-1)$, this residual divisor is integral and reduced: its expected codimension is one and $L(c-1) - 3 > 1$.

For completeness, these local product descriptions imply the global assertion. Their projections are flat, with geometrically integral fibres. Every irreducible component dominates B , whereas the geometric generic fibre is irreducible, so there is just one component. The product descriptions are reduced. The section is nonzero and is locally a nonzerodivisor, hence defines a Cartier divisor. The fibres of $S_{\varepsilon} \rightarrow P$ are the affine torsors

$$\prod_{\nu} \text{Hom}(H_{\varepsilon_{\nu}}/\mathcal{R}, \mathcal{R}/\mathcal{W}_{\nu}).$$

Local splittings trivialize them. Pulling back Z_P is consequently locally a product with affine space, proving the last assertion by the same generic-fibre and reducedness arguments. $\square$

Proposition 12.5 (Intrinsic normal-derivative factorization). *Let $Y = \mathbb{P}(V_{2n}^*) \times X_L$, let $S = S_\varepsilon$ be the signed open incidence over B , and let $\mathcal{E} = \bigoplus_\nu \text{Sym}^2 \mathcal{U}_\nu$. The incidence equations are a section s of $\mathcal{E}^* \otimes \mathcal{L}^*$, where $\mathcal{L}$ is the annihilator line. Their normal derivative is a map of rank- e bundles*

$$d_N s : N_{S/Y} \longrightarrow \mathcal{E}^* \otimes \mathcal{L}^*.$$

After passing to the labelled etale chart where necessary, the normal bundle has an exact sequence

$$0 \longrightarrow \bigoplus_\nu \text{Hom}(\mathcal{U}_\nu, V_n/H_{\varepsilon_\nu}) \longrightarrow N_{S/Y} \longrightarrow \mathcal{L}^* \otimes \mathcal{T}^* \longrightarrow 0. \quad (12.7)$$

Put $Q = \bigoplus_\nu \text{Sym}^2 \mathcal{W}_\nu$. The first term maps isomorphically onto the kernel of the restriction $\mathcal{E}^* \otimes \mathcal{L}^* \rightarrow Q^* \otimes \mathcal{L}^*$. On the remaining associated quotients, $d_N s$ is $\text{id}_{\mathcal{L}^*} \otimes \tau^*$. Consequently its determinant and the residual determinant are identified by an isomorphism of their determinant line bundles. In any local frames this reads

$$\det(d_N s) = u_0 \det(\mu'_K), \quad u_0 \in \mathcal{O}_S^*. \quad (12.8)$$

The unit ambiguity is exactly the change of frames; the equality of zero divisors and their multiplicities is intrinsic.

Proof. First restrict Y to the rank-two base. Relative to this base the normal bundle of $\text{Gr}(c, H_{\varepsilon_\nu})$ in $\text{Gr}(c, V_n)$ is $\text{Hom}(\mathcal{U}_\nu, V_n/H_{\varepsilon_\nu})$. The normal sequence for $S \subset Y|_B \subset Y$ gives the first term of (12.7) and the normal bundle of B in $\mathbb{P}(V_{2n}^*)$ as quotient.

The affine tangent space of the rank-two cone at ℓ is spanned by the two evaluations and their first derivatives, with the latter weighted by the nonzero coefficients. A polynomial annihilating this four-dimensional space vanishes to order two at both support points. Its space is precisely $g^2 V_{2n-4}$. Therefore

$$N_{B/\mathbb{P}(V_{2n}^*)} = \text{Hom}(\mathcal{L}, \mathcal{T}^*) = \mathcal{L}^* \otimes \mathcal{T}^*.$$

This identification is independent of coordinates on $\mathbb{P}^1$.

For one contact write $U = W \oplus \mathbb{C}a$ locally. The restriction of a rank-two form has one nonzero quotient functional in the V_n/H_ε direction. Varying each vector of W in this direction prescribes the wa entries, and varying a prescribes the a^2 entry, with its nonzero factor two. The ww' entries remain zero. Thus the plane-normal map is an isomorphism onto the rank- c space of equation forms vanishing on $\text{Sym}^2 W$. Doing this for each contact proves the first block assertion.

Restrict the remaining equations to Q . Variations of the planes contribute zero there. An annihilator-normal variation $\dot{\ell}$ contributes $q \mapsto \dot{\ell}(\mu(q))$, exactly τ^* under the preceding normal-bundle identification. Splittings of the two exact sequences therefore give an upper block triangular matrix with an invertible plane block and the block $\text{id}_{\mathcal{L}^*} \otimes \tau^*$. Taking determinants proves the assertion on determinant lines. A new splitting alters only an off-diagonal block, and new frames change determinants by units. Finally trivializing $\mathcal{G}$ cancels the common g^2 factor and identifies τ with μ'_K , giving (12.8). $\square$

Lemma 12.6 (A simultaneous nonempty corank-one wall open). *The integral divisor Z_ε has a nonempty open subset where the residual map has corank one and nonsingular Hankel annihilator η , $K_1 \rightarrow \mathbb{C}^2$ has full evaluation rank at the support points, and the original multiplication has corank one. This open subset is dense in Z_ε .*

Proof. The first two residual conditions hold on a nonempty open subset of the integral residual divisor by Theorem 11.1. We verify that the evaluation-rank condition meets this subset rather than presuming generality of a residual plane.

Write $c' = c - 1$, $k' = k - 2$; here $k' \geq 2c' + 1$ and $c' \geq 9$. For a nondegenerate form on a k' -space, the locus of isotropic c' -planes contained in a fixed hyperplane has codimension at least c' in its orthogonal Grassmannian. For a nonisotropic normal vector this follows by replacing k' by $k' - 1$ in the dimension formula. For an isotropic normal vector v , planes not containing v are lifts of isotropic c' -planes in $v^\perp/\mathbb{C}v$, with c' affine lift parameters, giving the same bound. Planes containing v are parametrized by $\text{OGr}(c' - 1, k' - 2)$ and have codimension $k' - c' - 1 \geq c'$. This proves the bound in all cases.

Failure of K_1 to map onto the two evaluation coordinates means containment in one member of their hyperplane pencil. The union has codimension at least $c' - 1 > 0$. Over the nondegenerate residual annihilator incidence, its complement is thus nonempty and open. That incidence is irreducible and dominates the residual determinant divisor; its corank-one open is nonempty. Intersecting these two nonempty opens in an irreducible space proves the simultaneous residual assertion, for each fixed pair of support points.

Choose such a tuple and a nonzero $k_0 \in K_1 \cap gV_{n-4}$; this intersection has dimension at least $c - 3 > 0$. Write $k_0 = gk'_0$ and $U_1 = gK_1 + \mathbb{C}a_1$. The already spanned space $g^2 \text{im } \mu'_K$ has one missing direction in g^2V_{2n-4} . The product $(gk_0)a_1 = g^2k'_0a_1$ supplies it if $\eta(k'_0a_1) \neq 0$. Changing the lift by $a_1 \mapsto a_1 + gw$ changes this value by $H_\eta(k_0, w)$. This is a nonzero functional because η is nondegenerate. It vanishes on K_1 , since K_1 is isotropic, so it is a well-defined nonconstant affine function on the actual lift torsor, not merely on a space of redundant frames.

After this choice, products from $(gK_1)a_1$ supply both independent first-derivative coordinates modulo g^2V_{2n-4} , using full evaluation rank and the nonzero values of a_1 at both support points. The square a_1^2 supplies a further, nonzero value coordinate. The product span therefore has dimension at least $(p - 4) + 3 = p - 1$. It is annihilated by ℓ , so its dimension is exactly $p - 1$. All the conditions used are open, and we have exhibited choices meeting them. Integrality of Z_ε from Proposition 12.4 proves density. $\square$

12.4. A simple normal determinant gives an exact double crossing. The passage from a normal derivative to a local equation is an independent analytic statement. We record the precise hypotheses; first-order corank alone would not suffice.

Lemma 12.7 (Simple normal determinant). *Let Y be a smooth complex variety of dimension $d + e$, let $S \subset Y$ be smooth of codimension e , and let s be a section of a rank- e vector bundle $\mathcal{F}$ which vanishes identically on S . At $z \in S$, assume that the normal derivative*

$$d_N s : N_{S/Y} \longrightarrow \mathcal{F}|_S$$

*has rank $e - 1$. Suppose its determinant has a simple zero on S at z , meaning that a local representative has nonzero differential in T_z^*S . Then $d \geq 1$ and the zero scheme $Z = V(s)$ has analytic local equation $uv = 0$ in a smooth $(d + 1)$ -dimensional space. More precisely,*

$$\hat{\mathcal{O}}_{Z,z} \simeq \mathbb{C}[[t_1, \dots, t_{d-1}, u, v]]/(uv), \quad (12.9)$$

with S corresponding to the branch $u = 0$. No reducedness or component-count assumption on Z is needed.

Proof. Choose coordinates $(t_1, \dots, t_d, z_1, \dots, z_e)$ in which $S = \{z_1 = \dots = z_e = 0\}$, and trivialize $\mathcal{F}$. After invertible changes of the normal coordinates and of the section's components, the first $e - 1$ components have an invertible Jacobian in $z_1, \dots, z_{e-1}$ at the point. The implicit function theorem solves these components as $z_i = h_i(t, u)$, where $u = z_e$. Since the components vanish on S , uniqueness implies $h_i(t, 0) = 0$. Their common zero scheme is a smooth $(d + 1)$ -dimensional neighbourhood containing $S = \{u = 0\}$. For $e = 1$ this step has no equations to eliminate.

Restrict the last component to this neighbourhood and call it $f(t, u)$. It vanishes on $u = 0$, so in the convergent power-series ring it factors exactly as $f = uG(t, u)$. Along S , the chain rule for

the eliminated equations gives

$$G(t, 0) = \partial_u f(t, 0) = \frac{\det(d_N s)(t)}{\det(\partial(s_1, \dots, s_{e-1})/\partial(z_1, \dots, z_{e-1}))(t, 0)}.$$

For $e = 1$ the denominator is 1. This is the elementary Schur complement identity for the normal Jacobian. The denominator is a unit. Since the numerator vanishes simply on S , the differential of $G(t, 0)$ is nonzero. Consequently du and dG are linearly independent at z . Complete $u, v = G(t, u)$ to analytic coordinates $(u, v, t'_1, \dots, t'_{d-1})$ on the neighbourhood. In these coordinates the final equation is exactly $uv = 0$. Completion proves (12.9). Changing any trivialization multiplies the determinants by units, so the simple-zero condition and the conclusion are intrinsic. $\square$

Corollary 12.8 (Normalization and conductor of the local crossing). *In Lemma 12.7, the normalization of the completed local ring is*

$$\mathbb{C}[[t_1, \dots, t_{d-1}, u]] \oplus \mathbb{C}[[t_1, \dots, t_{d-1}, v]],$$

and its conductor in that normalization is $(u) \oplus (v)$. The conductor in the original ring is (u, v) , and the scheme-theoretic intersection of the two branches is reduced and smooth of dimension $d - 1$.

Proof. The ring in (12.9) embeds into the displayed direct sum as the pairs whose restrictions at $u = v = 0$ agree. The sum is finite over it: adjoining the idempotent $(1, 0)$ suffices. Both summands are regular and integrally closed, so this is the normalization. A pair multiplies every pair in the sum back into the original ring precisely when both restrictions at zero vanish. This is $(u) \oplus (v)$, whose inverse image is (u, v) . The quotient by this ideal is $\mathbb{C}[[t_1, \dots, t_{d-1}]]$. $\square$

In the binary wall calculation below, Y is the annihilator ambient space, S the chosen signed incidence, and s its isotropy section. The relative determinant proposition identifies $\det(d_N s)$, up to a unit, with the residual multiplication determinant. The relative reduced divisor and the nonemptiness proposition supply its simple-zero locus while the original multiplication still has corank one. Lemma 12.7 then proves the exact local equation used in Theorem 12.9. Thus neither reducedness inferred from a tangent count nor an unproved removal of higher-order terms enters the crossing argument.

12.5. The local normal form.

Theorem 12.9 (The residual double-crossing divisor). *Under (12.5), Z_ε is irreducible and reduced. It has a nonempty dense open subset on which the following assertions hold. The original multiplication has corank one, and the residual multiplication has corank one with a nonsingular residual Hankel annihilator. Put $d = M - a$. The analytic germ of $\mathcal{D}_L$ at its projected point has equation $uv = 0$ in smooth local coordinates $(t_1, \dots, t_{d-1}, u, v)$, and its completed local ring is*

$$\mathbb{C}[[t_1, \dots, t_{d-1}, u, v]]/(uv). \quad (12.10)$$

The two branches are Σ_ε and Φ_L ; no other signed branch passes through this open subset. Their intersection has codimension one in either branch, and codimension $a + 1$ in X_L . The tangent cone is the union of two distinct hyperplanes in the $(d + 1)$ -dimensional tangent space. The germ is seminormal, its normalization separates its two smooth branches, and its conductor in (12.10) is (u, v) .

Proof. Proposition 12.4 gives the relative integral divisor, Lemma 12.6 supplies a simultaneous dense open, and Proposition 12.5 gives the intrinsic determinant identity. We spell out how they lead to the local equation. At residual dimensions $k' = k - 2$, $c' = c - 1$ the expected codimension is one and the signed codimension is $L(c - 1) - 3 > 1$. Here $c' \geq 9$ and $k' \geq 2c' + 1$. The already proved Theorem 11.1 says that the residual determinant is an integral reduced divisor whose

general annihilator η is nonsingular. Pullback by the smooth affine-space bundle in Proposition 9.3 preserves this generic irreducibility and reducedness; bundle trivializations as the support points vary give the same assertion for Z_ε .

We check that the original map has corank one on a nonempty open part of this divisor. A general residual tuple has K_1 mapping onto the two evaluation coordinates at x, y . Indeed failure of this condition places K_1 in one hyperplane of a pencil, and the isotropic Grassmannian incidence count excludes that closed subset. Choose a nonzero

$$k_0 \in K_1 \cap gV_{n-4}, \quad k_0 = gk'_0.$$

Such vectors exist because this intersection has dimension at least $c-3 > 0$. Write $U_1 = gK_1 + \mathbb{C}a_1$. The product $(gk_0)a_1 = g^2k'_0a_1$ fills the missing direction in g^2V_{2n-4} exactly when $\eta(k'_0a_1) \neq 0$. Varying the lift by $a_1 \mapsto a_1 + gw$, $w \in V_{n-2}$, changes this value by $H_\eta(k_0, w)$, a nonzero functional because H_η is nonsingular. Thus general lifts fill this direction. The products $(gK_1)a_1$ give both first-derivative coordinates modulo g^2V_{2n-4} , since K_1 has full evaluation rank and a_1 has nonzero values at x, y . The product a_1^2 gives one additional value coordinate. Consequently the original product span has dimension $(p-4) + 3 = p-1$. All these conditions are nonempty open conditions.

Work on this corank-one open set in the incidence. The signed incidence S is smooth of dimension d in the smooth ambient $Y = \mathbb{P}(V_{2n}^*) \times X_L$. Its codimension is

$$(M + p - 1) - d = p - 1 + b = e,$$

and the full incidence is the zero scheme of e restriction equations. The normal derivative of these equations along S splits into the Lc mixed and square equations involving the nonradical vector a_ν , and the $L\binom{c}{2} = p-4$ equations on $gK_\nu \times gK_\nu$. The first block is invertible in transverse plane coordinates, by the dual-frame proof of Theorem 9.1. The normal space to the rank-two Hankel base is identified with $(g^2V_{2n-4})^*$: its tangent space is the annihilator of g^2V_{2n-4} modulo $\mathbb{C}\ell$, as is seen by differentiating the two evaluations and their weights. In these normal coordinates the remaining derivative block is precisely $(\mu'_K)^*$. Hence, up to a nowhere-zero factor,

$$\det(\text{normal derivative along } S) = \det \mu'_K. \quad (12.11)$$

At a general point of Z_ε this derivative has corank one and its determinant vanishes simply along S .

Eliminate $e-1$ equations and their normal coordinates by the analytic implicit function theorem. The remaining ambient germ is smooth of dimension $d+1$, and S is the smooth hypersurface $u=0$ in it. The last equation vanishes on S , so it is uG . Its normal derivative along S is $G|_S$; by (12.11) this has a simple zero along the residual divisor. Thus dG has a nonzero component tangent to S , and $u, v = G$ are independent analytic coordinates. This proves (12.10), not only an equality of tangent dimensions. The corank-one projection identifies this germ with the one in X_L .

Only one sign choice is possible locally because no U_ν is contained in the radical and the annihilator is unique. The other branch must therefore be Φ_L , by Theorem 11.1. All dimension and tangent-cone assertions follow from the displayed local ring. Its normalization is

$$\mathbb{C}[[t_1, \dots, t_{d-1}, u]] \oplus \mathbb{C}[[t_1, \dots, t_{d-1}, v]],$$

Replacing formal by convergent power series gives the analytic normalization. In either case the original ring consists of pairs with the same restriction at $u = v = 0$. This identifies the conductor as (u, v) and proves seminormality: an element of the total fraction ring whose square and cube belong to the original ring lies in each normal branch ring, and its two restrictions agree because their squares and cubes agree. $\square$

The theorem describes a specified irreducible divisor and its dense open double-crossing locus. It does not claim that every point of $\Phi_L \cap \Sigma_\epsilon$ has this type, or that higher-corank and colliding-support intersections are normal crossings.

Corollary 12.10 (A logarithmic conditioning law at a crossing). *At a real point of the double-crossing locus, with its two real branches, there are real analytic coordinates $(z_1, \dots, z_{a-1}, u, v, t)$ in X_L in which*

$$\gamma_{\text{mult}} \asymp \left(\sum_{j=1}^{a-1} z_j^2 + u^2 v^2 \right)^{1/2}. \quad (12.12)$$

On a sufficiently small product box, a probability law whose density is bounded above and below by positive constants satisfies

$$\Pr\{\gamma_{\text{mult}} \leq \epsilon\} \asymp \epsilon^a \log(1/\epsilon), \quad \epsilon \downarrow 0. \quad (12.13)$$

The same law holds in any real loading chart submersing onto this neighbourhood. Constants are local and depend on the chosen chart and density bounds.

Proof. At corank one bounded invertible row and column operations compare γ_{mult} with the norm of the Schur-complement row. The analytic argument in Theorem 12.9, applied in the ambient Grassmannian chart, makes its ideal $(z_1, \dots, z_{a-1}, uv)$. Each finite set of generators is a bounded analytic linear combination of the other on a smaller compact chart, so their norms are comparable. This proves (12.12). For the upper bound integrate first over the $a - 1$ normal coordinates, whose volume is $O(\epsilon^{a-1})$. In a fixed square the area of $|uv| \leq C\epsilon$ is $O(\epsilon \log(1/\epsilon))$, by integrating $\min(r, C\epsilon/|u|)$ in u . For the lower bound restrict the normal coordinates to a ball of radius $\epsilon/2$ and take $|uv| \leq \epsilon/2$ in a smaller fixed square. The same elementary integral is bounded below by a positive constant times $\epsilon \log(1/\epsilon)$. Bounded density and Jacobian factors give the assertion. A submersion adds unconstrained smooth coordinates and does not change the exponent or logarithm. $\square$

This is different from the smooth-stratum law ϵ^a . It is a conditioning statement for a specified real design law, not a claim that the exact count-score compression experiment has Fisher information equal to an unwhitened squared singular value. The covariance of that score compression also depends on the design.

13. HIGHER PRODUCTS AND DIHEDRAL PHASE SECTORS

The general residual theorem of Section 6 already applies to every symmetric power. Here we calculate explicitly one distinguished family of annihilators within that general setting. The two-hyperplane description has a higher-product counterpart, but its monodromy is no longer exhausted by a simultaneous sign change. For $m \geq 2$ consider

$$\mu_m : \bigoplus_{\nu=1}^L \text{Sym}^m U_\nu \longrightarrow V_{mn}, \quad p_m = mn + 1, \quad e_m = L \binom{c + m - 1}{m}.$$

Here “two-point annihilator” refers to the family

$$[\ell] = [\text{ev}_x^{(mn)} - t^m \text{ev}_y^{(mn)}], \quad x \neq y, \quad t \neq 0,$$

not to a purported classification of every higher catalecticant rank. Let μ_m be the group of m th roots of unity.

Proposition 13.1 (Algebraic splitting cover and phase descent). *On the distinct-support two-point base for m th multiplication, the irreducible components of the reduced isotropic incidence are exactly the orbits of $\mu_m \rtimes \{1, \sigma\}$ on μ_m^L , where simultaneous multiplication acts on all labels and σ inverts all labels. No further splitting occurs for an inversion-fixed relative pattern.*

Proof. It suffices first to work over the dense chart where both support points are finite. Its labelled splitting cover is the integral variety

$$T = \{(x, y, t) \in \mathbb{A}^2 \times \mathbb{G}_m : x \neq y\}.$$

The transformations

$$\tau_\zeta(x, y, t) = (x, y, \zeta t), \quad \sigma(x, y, t) = (y, x, t^{-1})$$

satisfy $\sigma\tau_\zeta\sigma = \tau_{\zeta^{-1}}$. They form a group of order $2m$, acting freely: a nontrivial rotation cannot fix $t \neq 0$, and a reflection interchanges the distinct points. The quotient is a finite etale Galois cover of the two-point base. To identify the quotient with that base, note that the annihilator $[\text{ev}_x - t^m \text{ev}_y]$ recovers its unordered support and weight ratio. For example the radical of its middle Hankel pairing recovers the degree-two support polynomial, and a two-column Vandermonde system recovers the weights. This also gives algebraic inverses on the usual nonvanishing-minor charts. For the degrees under consideration the pairing has both side dimensions at least two. Thus the displayed transformations are the entire deck group, not just selected monodromies.

Over T the hyperplanes H_ζ are labelled. Their reduced incidences are the integral smooth bundles $\prod_\nu \text{Gr}(c, H_{\zeta_\nu})$; they have geometrically integral fibres. A deck transformation permutes these bundles by the displayed action on the labels (using the inverse rotation gives the same orbit action). The image of any integral bundle under the finite quotient map is irreducible and is exactly the image of its orbit union. Different orbits have distinct generic geometric components. Hence these images are precisely the irreducible components downstairs. An element stabilizing one pattern acts on its integral bundle; its quotient remains integral, since an invariant subring of a domain is a domain. In particular a fixed relative inversion pattern does not split after descent.

Changing affine coordinates on $\mathbb{P}^1$ only changes trivializations of the evaluation lines. Such charts cover the base, and their nonempty overlaps give the same hyperplane-label cover. Alternatively, each component of the full incidence dominates the base, because its split local models are Grassmannian bundles. Thus it is the closure of a component on this dense chart. No additional component is hidden at infinity. This completes the algebraic descent proof. $\square$

Theorem 13.2 (Phase-sector classification for every symmetric power). *Suppose $n \geq 4$, $4 \leq c < n + 1$, $L \geq 1$, and $m \geq 2$. The reduced incidence of two-point annihilators of μ_m has*

$$N_{m,L} = \frac{m^{L-1} + \gcd(m, 2)^{L-1}}{2} \quad (13.1)$$

irreducible components on the distinct-support open base. The closure of the image of each component in X_L is irreducible and rational of codimension $Lc - 3$. Its incidence projects birationally, and its general point has a unique two-point annihilator. These sectors are indexed by relative phase patterns

$$(\xi_2, \dots, \xi_L) \in \boldsymbol{\mu}_m^{L-1}, \quad (\xi_2, \dots, \xi_L) \sim (\xi_2^{-1}, \dots, \xi_L^{-1}).$$

For $m = 2$ this recovers 2^{L-1} . In general the count is not m^{L-1} .

Proof. By polarization in characteristic zero, ℓ annihilates $\text{Sym}^m U$ if and only if the degree- m polynomial $\ell(f^m)$ vanishes identically for $f \in U$. Its factorization is

$$f(x)^m - t^m f(y)^m = \prod_{\zeta \in \boldsymbol{\mu}_m} (f(x) - \zeta t f(y)).$$

Since the polynomial ring on U is a domain, this holds precisely when U lies in one of the m hyperplanes $H_\zeta = \ker(\text{ev}_x - \zeta t \text{ev}_y)$. The splitting cover of the two-point base labels all these hyperplanes. Changing the chosen m th root t acts by simultaneous cyclic translation of phase labels. Interchanging the two support points acts by inversion after that translation. Proposition 13.1 constructs the finite etale cover, identifies its full deck group, and proves the

descent assertion, including sectors with stabilizers. Thus the component orbits are those of the dihedral action on μ_m^L .

Quotient first by simultaneous translation, normalizing the first phase to one. The remaining involution inverts every relative phase. Its fixed patterns have each entry of order dividing two; there are $\gcd(m, 2)^{L-1}$ of them. The two-element orbit formula gives (13.1). Over the splitting cover every phase sector is a smooth irreducible product of relative Grassmannians; monodromy orbits are therefore exactly the irreducible components after descent.

Each incidence has dimension $3 + Lc(k - 1 - c) = M - (Lc - 3)$. The first plane determines its general secant hyperplane uniquely by the dimension argument in Lemma 11.4. That hyperplane determines its unordered support points, weight ratio and resulting two-point annihilator, and the other planes determine their phase hyperplanes. This proves generic injectivity and distinctness of the image sectors, hence birationality in characteristic zero.

For rationality, a relative pattern not fixed by inversion chooses an ordering of the support points. Its parameterization uses an open subset of $\mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{G}_m$, with rational Grassmannian fibres trivialized on a dense open set, and is birational. For an inversion-fixed pattern, its first hyperplane instead parameterizes the unordered secant threefold, which is rational; all other hyperplanes are rationally determined from it. The same Grassmannian argument applies. Taking image closures does not change these birational assertions. $\square$

The theorem classifies the two-point incidence and its image sectors. It does not assert that every such sector is an irreducible component of the entire higher-product failure scheme, or that the full higher-product failure codimension has been determined.

Theorem 13.3 (Higher-product residual conormal). *On a phase sector suppose no U_ν is contained in $E = gV_{n-2}$, and write $U_\nu = gK_\nu + \mathbb{C}a_\nu$. Put*

$$R_m = \sum_{\nu} (gK_{\nu})^2 U_{\nu}^{m-2} = g^2 \sum_{\nu} K_{\nu}^2 U_{\nu}^{m-2} \subset V_{mn}, \quad \delta_m = (p_m - 4) - \dim R_m.$$

If $e_m \geq p_m$ and μ_m has corank one, then the tangent dimension of its maximal-minor scheme is

$$M - (Lc - 3) + \delta_m. \tag{13.2}$$

In particular $\delta_m = 0$ implies scheme smoothness of codimension $Lc - 3$ at this point.

Proof. For a fixed contact split $U = W + \mathbb{C}a$, with $W = gK$. The derivative with respect to the plane vanishes on every monomial having at least two W factors: after varying one factor another still vanishes at both support points. On the quotient spanned by a^m and wa^{m-1} , $w \in W$, it is surjective. Indeed the nonzero quotient linear functional

$$v \mapsto v(x)a(x)^{m-1} - t^m v(y)a(y)^{m-1}$$

allows the coordinates of the variation of each basis vector to solve these c equations independently; the coefficient for a^m is the nonzero integer m . Thus the plane derivative has rank c . The remaining compatibility equations for ℓ are precisely annihilation of R_m . Independent solutions for the contacts give

$$\dim T\mathcal{J}_m = M + (p_m - 1) - Lc - \dim R_m,$$

which is (13.2). The corank-one Schur argument identifies this tangent space with that of the maximal-minor scheme. Multiplication by g^2 embeds V_{mn-4} , of dimension $p_m - 4$, so $\delta_m \geq 0$. If it is zero, the tangent dimension equals the dimension $M - (Lc - 3)$ of the local phase incidence, which is embedded by the corank-one projection. Equality of tangent and local dimensions proves smoothness. $\square$

For $m = 2$ the residual span is independent of the lifts a_ν and reduces to the multiplication of the K_ν . For $m > 2$ it retains U_ν^{m-2} . This difference is why a higher-product phase classification alone does not imply the quadratic saturation formula with q replaced by $\binom{c+m-1}{m}$.

14. A GLOBAL HIGHER-PRODUCT FAMILY WITH EMBEDDED STRUCTURE

We now apply the residual mechanism to a natural family in every symmetric degree, determining the whole reduced failure locus rather than only an obstruction subfamily. The scheme structure contains a further phenomenon: its support is independent of the degree, but its nilpotent structure is not.

Fix $n \geq 4$ and put $V_n = H^0(\mathbb{P}^1, \mathcal{O}(n))$. A hyperplane $U \subset V_n$ is the kernel of a unique point $[h] \in \mathbb{P}(V_n^*)$. For $m \geq 2$ let

$$\mu_m : \text{Sym}^m \mathcal{U} \longrightarrow V_{mn} \otimes \mathcal{O}_{\mathbb{P}(V_n^*)}, \quad D_m = V(I_{mn+1}(\mu_m)).$$

Its source rank is $e_m = \binom{n+m-1}{m} \geq mn + 1$. Write $C_n \subset \mathbb{P}(V_n^*)$ for the evaluation rational normal curve, and S_n for its secant threefold, including tangent secants. Let $\mathcal{J}_m$ be the ideal sheaf of D_m , and $\mathcal{I}_{S_n}, \mathcal{I}_{C_n}$ the reduced ideal sheaves. Associated points below are those of coherent sheaves on projective space; no assertion about an irrelevant-ideal component of an affine homogeneous presentation is implicit.

Theorem 14.1 (Higher-product hyperplanes: support, ranks and associated points). *For every $n \geq 4$ and $m \geq 2$,*

$$(D_m)_{\text{red}} = S_n, \quad \text{corank } \mu_{m, \ker h} = \begin{cases} 0, & [h] \notin S_n, \\ 1, & [h] \in S_n \setminus C_n, \\ m, & [h] \in C_n. \end{cases} \quad (14.1)$$

The scheme D_m is smooth of dimension three at every point of $S_n \setminus C_n$, including the non-evaluation tangent secants. In particular

$$\mathcal{J}_m : \mathcal{I}_{C_n}^\infty = \mathcal{I}_{S_n}. \quad (14.2)$$

For every $m \geq 3$, the complete set of associated points is

$$\text{Ass}_{\mathcal{O}_{\mathbb{P}(V_n^*)}}(\mathcal{O}_{D_m}) = \{\eta_{S_n}, \eta_{C_n}\}. \quad (14.3)$$

Thus there is exactly one reduced irreducible component, and exactly one embedded associated subvariety, the evaluation curve. Its nilradical $\mathcal{N}_m = \mathcal{I}_{S_n}/\mathcal{J}_m$ satisfies, at every closed point $x \in C_n$,

$$(\mathcal{N}_m^j)_x \neq 0 \quad \text{whenever } 1 \leq j \text{ and } 2j < m. \quad (14.4)$$

Consequently the nilpotency index at x is at least $\lceil m/2 \rceil$. The theorem does not assert that this lower bound is the exact index.

Lemma 14.2 (Propagation from quadratic surjectivity). *Let $U \subset V_n$ be base-point-free. Then $UV_d = V_{n+d}$ for every $d \geq n - 1$. In particular, $U^2 = V_{2n}$ implies $U^m = V_{mn}$ for every $m \geq 2$.*

Proof. There are coprime $f, g \in U$: fix a nonzero f and choose g outside the finitely many proper evaluation hyperplanes at its zeros. The kernel of

$$V_d \oplus V_d \longrightarrow V_{n+d}, \quad (A, B) \longmapsto fA + gB$$

consists exactly of $(gh, -fh)$ with $h \in V_{d-n}$. For $d = n - 1$ its kernel is zero and both dimensions are $2n$. For $d \geq n$ the rank is $2(d + 1) - (d - n + 1) = d + n + 1$. Thus the map is onto in both cases. If $U^2 = V_{2n}$ then U is automatically base-point-free; apply the first assertion successively with $d = 2n, 3n, \dots$ $\square$

Lemma 14.3 (All higher-product ranks on secants). *For a non-evaluation point $h \in S_n$, the image U^m has codimension one in V_{mn} for every $m \geq 2$. For an evaluation point it has codimension m .*

Proof. First take a distinct-support secant and choose the two points as 0 and ∞ . In the binary monomial basis $z_j = X^j Y^{n-j}$, its hyperplane has a basis

$$z_1, \dots, z_{n-1}, \quad a = z_0 + \lambda z_n, \quad \lambda \neq 0.$$

Products of two of the interior monomials give all exponents $2, \dots, 2n-2$. Multiplication of a by z_1 and z_{n-1} then supplies exponents 1 and $2n-1$, since the other terms are already in that span. Finally a^2 gives the one allowed combination of the endpoint monomials. Thus U^2 contains every interior monomial and one endpoint combination, and has dimension $2n$.

Inductively suppose U^m contains all monomials of exponents $1, \dots, mn-1$, as well as a^m . Multiplying its interior monomials by $z_1, \dots, z_{n-1}$ gives exponents $2, \dots, (m+1)n-2$. Multiplying its two extreme interior monomials by a supplies the two missing interior exponents. The element a^{m+1} supplies an endpoint combination. All products satisfy the single endpoint relation inherited from $f(0) = \lambda^{-1}f(\infty)$ for $f \in U$. Hence there can be no second endpoint direction, and the image has dimension $(m+1)n$.

At a non-evaluation tangent secant, a change of homogeneous coordinates puts the hyperplane in the form with the first derivative at 0 equal to zero. To justify the normalization, in a local coordinate a change fixing 0 changes a first-jet functional by a nonzero scalar and an arbitrary multiple of evaluation; the latter multiple can therefore be removed. The hyperplane is then

$$U = \text{span}\{z_0, z_2, z_3, \dots, z_n\}.$$

Sums of m exponents in $\{0, 2, 3, \dots, n\}$ give exactly $\{0, 2, 3, \dots, mn\}$. This follows by induction, since the intervals obtained by adding $[2, n]$ overlap or are adjacent for $n \geq 3$. Thus precisely exponent 1 is missing.

Finally, for h an evaluation, $U = lV_{n-1}$ for its vanishing linear form l . Since products of the complete binary series span, $U^m = l^m V_{m(n-1)}$, of dimension $m(n-1) + 1$. Its corank is m . $\square$

Lemma 14.4 (Explicit plane-derivative matrix and residual span). *Use the monomials $z_j = X^j Y^{n-j}$. At a split secant take*

$$a = z_0 + \lambda z_n, \quad (w_1, \dots, w_{n-1}) = (z_1, \dots, z_{n-1}), \quad v = z_n, \quad \lambda \neq 0,$$

and let the annihilator be coefficient mn minus λ^m times coefficient zero. At a tangent secant take $a = z_0$, $(w_1, \dots, w_{n-1}) = (z_2, \dots, z_n)$, $v = z_1$, and the coefficient-one annihilator. Set $u_0 = a$, $u_i = w_i$, and vary u_i by $t_i v$. On the ordered source quotient

$$a^m, w_1 a^{m-1}, \dots, w_{n-1} a^{m-1}$$

the plane-derivative matrix is

$$\kappa \text{ diag}(m, 1, \dots, 1), \quad \kappa = \lambda^{m-1} \text{ or } 1, \quad (14.5)$$

respectively. It is zero on all monomials with at least two w factors. Their product image has dimension $mn-3$ in both cases.

Proof. Let ℓ be the indicated annihilator. Then $\ell(va^{m-1}) = \kappa \neq 0$. The variation of a^m gives $m\kappa t_0$. For $w_i a^{m-1}$ only variation of w_i can contribute to ℓ , giving κt_i ; all terms which retain w_i vanish. For a monomial with at least two w factors one such factor remains after every single variation, and the result is annihilated by ℓ . This gives every entry of (14.5).

In the split case $W = XYV_{n-2}$, so $W^2 = (XY)^2 V_{2n-4}$. The subseries $U = W + \mathbb{C}a$ is base-point-free: W has only the two endpoint base points, and a is nonzero at both. Since $2n-4 \geq n-1$ for $n \geq 4$, Lemma 14.2 can be applied successively at degrees $2n-4, 3n-4, \dots$, proving $W^2 U^{m-2} = (XY)^2 V_{mn-4}$.

In the tangent case the exponent set of W^2 is the entire integer interval $[4, 2n]$. If the exponent set at degree k is $[4, kn]$, adding the exponent set $\{0\} \cup [2, n]$ of U gives $[4, kn] \cup [6, (k+1)n] = [4, (k+1)n]$, since $kn \geq 5$. Induction from $k = 2$ proves the assertion in every degree. There are exactly $mn - 3$ exponents in $[4, mn]$. $\square$

Lemma 14.5 (Scheme smoothness off the evaluation curve). *At every point of $S_n \setminus C_n$, the plane derivative of the annihilator incidence has rank n , and its polar residual image has dimension $mn - 3$. Consequently the tangent space of D_m has dimension three at each such point.*

Proof. For distinct support, let g be its vanishing quadratic and write $U = W \oplus \mathbb{C}a$, with $W = gV_{n-2}$. The plane derivative vanishes on monomials with at least two factors in W , while it is an isomorphism onto the dual of the n -dimensional quotient spanned by a^m and wa^{m-1} . Indeed the functional on V_n/U obtained by varying one factor and evaluating against the annihilator is nonzero; the images of the n basis vectors of U can be varied independently. This is the computation of Theorem 13.3, now with $c = n$. The residual image is

$$W^2U^{m-2} = g^2V_{2n-4}U^{m-2} = g^2V_{mn-4}.$$

The last equality follows from Lemma 14.2, because U is base-point-free and $2n - 4 \geq n - 1$. Its dimension is $mn - 3$.

For a tangent secant in the normal form of Lemma 14.3, let $W = \text{span}\{z_2, \dots, z_n\}$ and $a = z_0$. The annihilator in V_{mn}^* is the coefficient of exponent 1. Since V_n/U is spanned by z_1 , the derivative independently prescribes the coefficients dual to a^m and wa^{m-1} , and has rank n . Once again its kernel on the dual source consists of monomials with at least two W factors. Their products span precisely the monomials of exponents $4, \dots, mn$: W^2 gives the full interval $[4, 2n]$, and successive factors from U extend it without gaps. This image also has dimension $mn - 3$.

The incidence tangent sequence now gives, in both cases,

$$\dim TD_m = \dim T\mathcal{I} = n + (mn + 1) - 1 - n - (mn - 3) = 3,$$

because the original multiplication has corank one by Lemma 14.3. $\square$

Lemma 14.6 (The length-two radical, including a double point). *Let $0 \neq \ell \in V_{2n}^*$ and let $H_\ell(f, g) = \ell(fg)$ on V_n . If H_ℓ has rank two, there is a length-two divisor $Z \subset \mathbb{P}^1$ such that*

$$\ell \in H^0(\mathcal{O}_Z(2n))^* \subset V_{2n}^*, \quad \text{rad } H_\ell = H^0(\mathcal{O}(n)(-Z)).$$

The induced pairing on $H^0(\mathcal{O}_Z(n))$ is nondegenerate.

Proof. The rank-at-most-two Hankel variety is the secant variety of the evaluation rational normal curve. The shape-independence of the Hankel minor ideal and the precise secant parametrization are given in [7, Section 1 and Section 2, equation (2.0.1)]; here the minor size is three. The projective bundle of the spaces $H^0(\mathcal{O}_Z(2n))^*$ over $\text{Hilb}^2 \mathbb{P}^1$ is proper. Its image contains the distinct-point secant lines and is contained in the rank-at-most-two locus by factorization through a two-dimensional restriction space. Its image is therefore the entire secant variety, including tangent lines. Thus ℓ factors through some Z .

For two distinct points the pairing on the length-two algebra has matrix $\text{diag}(\alpha, \beta)$. Rank two forces $\alpha\beta \neq 0$. For $Z = 2p$, choose a parameter z and trivialize the line bundle. Write $\ell(A + Bz) = \alpha A + \beta B$ on $\mathbb{C}[z]/(z^2)$. The pairing matrix in the basis $1, z$ is $\begin{pmatrix} \alpha & \beta \\ \beta & 0 \end{pmatrix}$, which has rank two exactly when $\beta \neq 0$. Restriction $V_n \rightarrow H^0(\mathcal{O}_Z(n))$ is surjective. In either case the kernel of the pulled-back nondegenerate pairing is exactly its restriction kernel $H^0(\mathcal{O}(n)(-Z))$. $\square$

Lemma 14.7 (Equivariant associated points on a transitive curve). *Let a group act on a Noetherian variety, preserving an ideal $\mathcal{I}$ with irreducible reduced support S . Let $C \subset S$ be a*

closed irreducible curve on whose closed points the group acts transitively. Suppose $\mathcal{J} = \mathcal{I}_S$ away from C and the nonzero module $\mathcal{N} = \mathcal{I}_S/\mathcal{J}$ has support exactly C . Then

$$\text{Ass}(\mathcal{O}/\mathcal{J}) = \{\eta_S, \eta_C\}.$$

Proof. For a finite module over a Noetherian ring, associated primes are finite, minimal points of its support are associated, and the associated primes of a submodule are associated to the ambient module; see [28, Lemma 10.63.3, Lemma 10.63.5, and Proposition 10.63.6]. These assertions apply on a finite affine cover and give the corresponding coherent-sheaf statements. Associated points outside C consist only of η_S , since the quotient there is the structure sheaf of a reduced irreducible variety. An associated point with closure inside C is either η_C or a closed point. The finite associated set is preserved by the group action. Transitivity on the infinitely many closed points of C excludes the latter possibility. Finally η_C is a minimal point of the support of $\mathcal{N}$, hence is associated to $\mathcal{N}$ and therefore to $\mathcal{O}/\mathcal{J}$. The minimal point η_S is associated as well. This gives exactly the two asserted points. $\square$

The coordinate assertion in Lemma 14.5 is supplied by (14.5). The length-two assertion and the associated-point step of the following proof are, respectively, Lemma 14.6 and Lemma 14.7. We retain the full argument to specify where its nilpotence estimate is used.

Proof of Theorem 14.1. We first identify the quadratic support. A hyperplane U can be totally isotropic for a nonzero Hankel form H_ℓ only if that form has rank at most two: the maximal isotropic dimension of a rank- r form on an $(n+1)$ -space is $n+1 - \lceil r/2 \rceil$. A rank-one Hankel form gives an evaluation hyperplane. A rank-two Hankel form has radical $H^0(\mathcal{O}(n)(-Z))$ for a length-two divisor Z on $\mathbb{P}^1$. For clarity, this last assertion includes a double point: the proper family of secant lines indexed by length-two divisors covers the rank-at-most-two Hankel variety of Lemma 8.1; on a length-two quotient the rank-two pairing is nondegenerate, so its kernel is exactly the indicated space. Since an isotropic hyperplane contains this radical, its annihilating functional lies on the corresponding secant or tangent line to C_n . Hence $|D_2| \subset S_n$. Conversely a distinct-support secant imposes a nonzero quadratic endpoint relation. Closedness supplies the whole S_n , proving $|D_2| = S_n$.

Outside S_n , quadratic multiplication is onto; the propagation lemma proves surjectivity in every higher degree. On S_n the rank lemma gives the two remaining cases of (14.1). Thus $|D_m| = S_n$ globally. Since S_n is an irreducible threefold, Lemma 14.5 equates local Krull and tangent dimensions at every point off C_n . The local rings there are regular. In particular $\mathcal{J}_m = \mathcal{I}_{S_n}$ away from C_n .

The coherent quotient $\mathcal{N}_m = \mathcal{I}_{S_n}/\mathcal{J}_m$ is therefore supported on C_n , and some power of $\mathcal{I}_{C_n}$ kills it. This gives the inclusion of $\mathcal{I}_{S_n}$ in the left side of (14.2). The reverse inclusion holds because $\mathcal{I}_{S_n}$ is prime and S_n is not contained in C_n . This proves the saturation identity.

At an evaluation point x , the original corank is m . Schur elimination consequently expresses $\mathcal{J}_{m,x}$ as the ideal of m -minors of a matrix whose entries vanish at x . In the regular local ring $\mathcal{O}_{\mathbb{P}(V_n^*),x}$, with maximal ideal $\mathfrak{m}_x$, one has

$$\mathcal{J}_{m,x} \subset \mathfrak{m}_x^m. \quad (14.6)$$

Choose coordinates $h_0, \dots, h_n$ dual to the binary monomials so that $x = [1 : 0 : \dots : 0]$, and work on $h_0 = 1$. The function

$$F = \det \begin{pmatrix} 1 & h_1 & h_2 \\ h_1 & h_2 & h_3 \\ h_2 & h_3 & h_4 \end{pmatrix}$$

vanishes on S_n , since every two-evaluation moment sequence has Hankel rank at most two. Its initial form at x is $h_2 h_4 - h_3^2$, a nonzero quadratic. Therefore F^j has order $2j$, and (14.6) implies

that its class modulo $\mathcal{J}_m$ is nonzero whenever $2j < m$. Transitivity of $\mathrm{SL}_2(\mathbb{C})$ on C_n proves (14.4) at every evaluation point. For $m \geq 3$ this shows in particular that $\mathcal{N}_m$ is nonzero with support all of C_n .

It remains to exclude any additional associated points. The multiplication and its maximal-minor ideal are $\mathrm{SL}_2(\mathbb{C})$ -equivariant. The associated points of a coherent sheaf form a finite set, stable under automorphisms. Off C_n the scheme is the smooth irreducible open part of S_n , so its only associated point there is η_{S_n} . An associated point whose closure is contained in C_n is either η_{C_n} or a closed point of C_n . The latter is impossible: transitivity would force every closed point of the curve into this finite set. Thus the only possible embedded associated point is η_{C_n} . It actually occurs because the nonzero submodule $\mathcal{N}_m$ has support C_n , whose generic point is a minimal point of its support and hence an associated point. Associated points of a submodule are associated points of the ambient module. This proves (14.3), and completes the proof. $\square$

The expected determinantal codimension here is $\binom{n+m-1}{m} - mn$, while the actual codimension is $n - 3$. For $m \geq 3$ this is an excess family with a complete associated-point classification. The nilpotency bound is unbounded as m grows even though the reduced threefold and its unique embedded support do not change. This conclusion is stronger than the two-point-sector calculation: it excludes every other reduced failure component and every other embedded associated subvariety in this entire family. It does not identify the full embedded primary ideal, or assert an all-contact, all- c higher-product classification.

15. COMPARISON WITH PRIOR MULTIPLICATION AND FAILURE-LOCUS RESULTS

The use of secants to account for failure of normal generation is classical. It is important here to distinguish the fixed complete series, its varying proper subseries, the Hankel annihilator space, and the scheme on the parameter space of subseries. The following comparisons refer to specific results, not to an inference from the titles of articles.

Result compared	Parameter space and conclusion	Use and distinction here
[7, Section 1; Theorem 2.1]	Hankel matrices of indeterminates; determinantal dimensions, normality and rational singularities.	These supply the annihilator-rank geometry. They do not identify the singularities of its varying isotropic-plane image. Our residual Fitting formulas concern that image.
[8], [19, Section 2.9]	A matrix satisfying the expected-grade condition has its maximal-minor resolution.	We use this resolution after proving the height. The rank competition, exceptional incidences and residual normal derivative are separate arguments.
[5, Theorem 0.2; Proposition 2.2]	A fixed completely embedded surface; numerical cohomology conditions force failure on finite linear sections, including a higher-product version.	Here the subseries vary, without cross-products. These cited statements do not give the component or residual Fitting formulas on $\mathrm{Gr}(c, V_n)^L$.
[14, Theorems 1 and 3]	Complete very ample curve series; criteria and secant explanations for failure of normal generation.	The ambient complete series on $\mathbb{P}^1$ here is normally generated. The failure comes from varying proper subseries.

For the component theorem the additional assertions are the full rank-incidence optimization, the five exceptional isotropic incidences, and descent of the signed families. For the global singularity theorem they are the identification of the polar residual space and the equality of the restricted cotangent Fitting ideals with its multiplication minors. The local node theorem further uses a relative determinant divisor and its normal derivative, not merely the dimensions of rank

strata. The cotangent presentation, Schur elimination and determinantal resolution themselves are standard algebraic tools and are not claimed as new constructions.

The closely related paper [4] requires a distinct qualification. The preceding source audit inspected its publisher's first page, which introduces higher-order embedding conditions and varying restriction maps to finite schemes. The remaining pages, containing the theorem statements needed for a complete comparison, were not available in the primary text retrieved for this revision. Accordingly no assertion that its theorems exclude or contain the results here is made. The comparison with this particular paper remains unverified; the table above is not a substitute for it. In particular a difference of parameter-space descriptions on the first page would not by itself establish a priority claim.

The local presentations in Section 7 use classical Schur elimination and the analytic implicit function theorem. The generic determinantal singularity, the quadratic cone $xz - y^2 = 0$, and the cubic cusp are not claimed as new singularities. The assertion specific to multiplication is the formula for the joint polar relations, its single-annihilator specialization, and the explicit residual obstruction and realization maps. A theorem-level comparison of these formulations with the failure-locus literature must distinguish an exact nonlinear presentation from the restricted Fitting identities; neither is a substitute for the unresolved comparison with [4].

15.1. The intrinsic derivative and classical determinantal geometry. The intrinsic map to $\text{Hom}(\ker \mu, \text{coker } \mu)$ is the normal derivative to a matrix rank stratum. This interpretation, and the tangent condition $\psi(\ker \varphi) \subset \text{im } \varphi$, are classical; see [10, Remark 1.18]. Invertible row and column operations preserve minor ideals [10, Lemma 1.3], and the generic rank-stratum germ is a smooth factor times a smaller determinantal germ [10, Section 1.3.2, equation (13)]. Essential nonsingularity is transversality to that stratum [10, Definition 2.11 and Proposition 2.12]. Fitting ideals are independent of the chosen presentation and commute with base change [25, Tags 07Z8 and 07ZA].

Accordingly, neither the existence of an intrinsic normal derivative nor the generic determinantal normal form is claimed as a new general theorem. Section 7 gives a self-contained application of these tools, identifies their dual obstruction through the actual multiplication polars, and separates the resulting exact nonlinear equations from their first-order Fitting data. The structural hypotheses are explicitly tested there; the generic realization example is not used as a substitute for a natural higher-product classification.

The latter role belongs to Theorem 14.1. It proves a global statement on the natural parameter space $\mathbb{P}(V_n^*)$, for every $n \geq 4$ and $m \geq 2$, rather than inferring a classification from a general transversality criterion. The base-point-free pencil argument and rank-two Hankel description are classical ingredients. The additional scheme conclusions proved here are smoothness at every non-evaluation secant, saturation off the evaluation curve, the complete associated-point set for $m \geq 3$, and a degree-dependent lower bound on nilpotency. They follow from the polar rank computation, the order of the Schur minors at an evaluation, and equivariance. They are not consequences of the generic matrix normal form alone. This comparison identifies the content requiring a priority assessment; it does not assert that an exhaustive search has established its historical originality.

15.2. Contact-algebra reduction and the scope of the new classification. Theorems 2.2–4.2 concern the entire relative generating Grassmannian $X_{d,2}^\circ$, including all multiplicity partitions of its length- d divisor. The subseries have codimension $d - 2$; d is arbitrary. For this family the result is a complete scheme theorem, not only a two-point annihilator sector: the component ideals, primary exponents, absence of embedded points, nilpotent layers, all coranks, and moving-divisor normalization are explicit. The marked length-two divisor in the normalization is recovered from

the failure scheme; it is not imposed on the parameter space as a restriction to one preselected obstruction sector.

The determinant of a power basis, the Vandermonde and confluent Vandermonde identities, and the change-of-basis formula for discriminants are elementary classical identities; no priority is claimed for those identities. Their proofs are included to fix all multiplicities, including at nonreduced contacts. The exact multiplication-cokernel transport in Theorem 2.2 is what permits these identities to describe the original maximal-minor scheme after arbitrary base change. The resulting primary decomposition is not inferred from an order-of-vanishing lower bound.

For a precise comparison, [5, Theorem 0.2 and Proposition 2.2] are existence results for deficient finite linear sections of a fixed completely embedded surface. Those statements do not contain the contact-pencil primary formula or the normalization constructed here. This observation is a comparison of the specified statements, not a claim about all work on failure loci. The theorem pages of [4] have still not been obtained, so its comparison remains explicitly undetermined. The accompanying source audit records the parameter-space, degree, scheme, corank, and associated-prime questions separately; a missing source is not treated as evidence of novelty.

16. AN APPLICATION TO CALIBRATED INFORMATION RECOVERY

Let J be fixed and invertible, let $H(\theta)_{ij} = \theta_{i+j}$, and set $R_\theta = JH(\theta)J^\top$ in its positive moment cone. For a prescribed smooth contact, a normal frame Z determines $U = J^\top \text{ran } Z$. The normal Hessian of the squared R_θ^{-1} -distance is the inverse of $Z^\top R_\theta Z$ (Lemma B.1). Hence this contact gives exactly the restriction of the moment functional to U^2 . Several independent contacts give the dual of (5.1). The information-loss locus is therefore the real positive realization of the failure geometry studied above, subject to the exact contact hypotheses.

Theorem D.1 supplies the positive native realization and global separation conditions. At a real crossing with two real branches, Corollary 12.10 gives the local small-singular-value law $\varepsilon^a \log(1/\varepsilon)$. These are applications of the geometry, not assumptions in its proofs. The appendices give the complete likelihood, boundary, score-compression, finite-offset and estimation arguments. In particular they do not identify a computed feature with an independent noisy distance observation. *Accompanying appendices.* All application and technical appendices cited in this article are reproduced without abridgement in *Information recovery and statistical appendices*. The complete archival manuscript combines both parts in one numbered document.

REFERENCES

- [1] R. Balan, P. Casazza, and D. Edidin, On signal reconstruction without phase, *Appl. Comput. Harmon. Anal.* **20** (2006), 345–356.
- [2] E. Ballico and Ph. Ellia, The maximal rank conjecture for nonspecial curves in $\mathbb{P}^3$, *Invent. Math.* **79** (1985), 541–555.
- [3] E. Ballico and Ph. Ellia, The maximal rank conjecture for non-special curves in $\mathbb{P}^n$, *Math. Z.* **196** (1987), 355–367.
- [4] E. Ballico, On the Failure Locus of Higher Order Properties of Embeddings in Projective Spaces, *Math. Nachr.* **163** (1993), 5–13, doi:10.1002/mana.19931630102.
- [5] E. Ballico, On the failure cycles for the quadratic normality of a projective variety, *Pacific J. Math.* **172** (1996), 307–313.
- [6] S. Basu, R. Pollack, and M.-F. Roy, *Algorithms in Real Algebraic Geometry*, second ed., Springer, 2006.
- [7] A. Conca, M. Mostafazadehfard, A. K. Singh, and M. Varbaro, Hankel determinantal rings have rational singularities, *Adv. Math.* **335** (2018), 111–129; arXiv:1709.05685, Sections 1–2.
- [8] J. A. Eagon and D. G. Northcott, Ideals defined by matrices and a certain complex associated with them, *Proc. Roy. Soc. London Ser. A* **269** (1962), 188–204.
- [9] D. Eisenbud, C. Huneke, and B. Ulrich, Heights of ideals of minors, *Amer. J. Math.* **126** (2004), 417–438; arXiv:math/0209184.

- [10] A. Frühbis-Krüger and M. Zach, Determinantal singularities, arXiv:2106.04855v1, 2021; Lemma 1.3, Remark 1.18, Section 1.3.2, and Proposition 2.12.
- [11] W. Fulton, *Intersection Theory*, second ed., Springer, 1998, Theorem 14.4.
- [12] I. M. Gelfand, M. M. Kapranov, and A. V. Zelevinsky, *Discriminants, Resultants, and Multidimensional Determinants*, Birkhäuser, 1994.
- [13] C. J. Geyer, On the asymptotics of constrained M -estimation, *Ann. Statist.* **22** (1994), 1993–2010.
- [14] M. Green and R. Lazarsfeld, On the projective normality of complete linear series on an algebraic curve, *Invent. Math.* **83** (1986), 73–90.
- [15] A. B. Hempel, P. J. Goulart, and J. Lygeros, Inverse parametric quadratic programming and an application to hybrid control, *IFAC Proceedings Volumes* **45**(17) (2012), 68–73.
- [16] G. Kempf and D. Laksov, The determinantal formula of Schubert calculus, *Acta Math.* **132** (1974), 153–162.
- [17] J. Kohonen, A meet-in-the-middle algorithm for finding extremal restricted additive 2-bases, *J. Integer Seq.* **17** (2014), Article 14.6.8.
- [18] J. Kohonen, An improved lower bound for finite additive 2-bases, *J. Number Theory* **174** (2017), 518–524.
- [19] A. R. Kustin, Canonical complexes associated to a matrix, arXiv:1509.04367, 2015, discussion following (1.0.2) and Section 2.9.
- [20] E. Larson, The maximal rank conjecture, arXiv:1711.04906, Theorem 1.2.
- [21] B. Mourrain and V. Y. Pan, Multivariate polynomials, duality, and structured matrices, *J. Complexity* **16** (2000), 110–180.
- [22] N. A. Nguyen, S. Olaru, P. Rodriguez-Ayerbe, M. Hovd, and I. Necoara, Constructive solution of inverse parametric linear/quadratic programming problems, *J. Optim. Theory Appl.* **172** (2017), 623–648.
- [23] C. Scheiderer, Extreme points of Gram spectrahedra of binary forms, *Discrete Comput. Geom.* **67** (2022), 1174–1190.
- [24] The Stacks Project Authors, *The Stacks Project*, Tag 031R, Lemma 10.157.3, the $R_0 + S_1$ reducedness criterion; consulted September 22, 2026.
- [25] The Stacks Project Authors, *The Stacks Project*, Section 15.8, Tags 07Z8 and 07ZA, presentation independence and base change of Fitting ideals; consulted September 22, 2026.
- [26] A. W. van der Vaart, *Asymptotic Statistics*, Cambridge University Press, 1998.
- [27] G. Yu, Upper bounds for finite additive 2-bases, *Proc. Amer. Math. Soc.* **137** (2009), 11–18.
- [28] The Stacks Project Authors, *The Stacks Project*, Section 10.63, Tag 00L9; Lemmas 10.63.3 and 10.63.5 and Proposition 10.63.6, associated primes; consulted September 22, 2026.